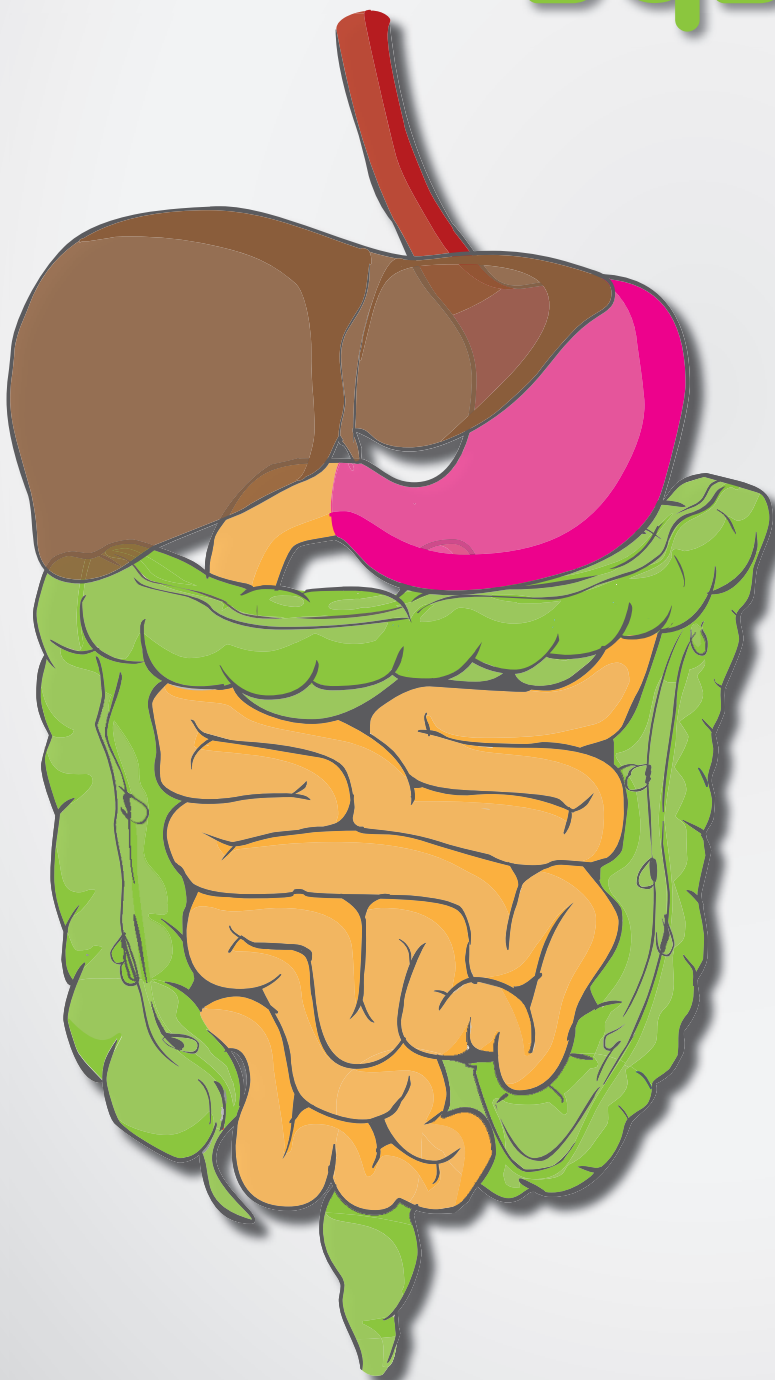


Memorix Anatomy

5

Digestive system



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The digestive system is comprised of the **digestive tract** extending from the mouth to the anus and of **glands** appended to the canal. Almost the entire digestive canal is built of **four layers**: mucosa, submucosa, a muscular layer and an outer layer that is either serosa or adventitia. The oral cavity, pharynx and oesophagus are described in their separate topographical spaces. The organs of the digestive system within the abdominal cavity are described by their relationship to the peritoneum as being either intra-peritoneal, primary retroperitoneal or secondary retroperitoneal. They are supplied by three unpaired trunks from the abdominal aorta: the coeliac trunk and the superior and inferior mesenteric arteries. Blood drains from the abdominal organs via the hepatic portal vein to the liver. Autonomic innervation of the abdominal organs is provided by the **vagus nerve** (parasympathetic) and the **splanchnic nerves** (sympathetic).

Basic parts

- 1 Oral cavity (*cavitas oris*)
- 2 Pharynx
- 3 Oesophagus
- 4 Stomach (*gaster*)
- 5 Small intestine (*intestinum tenue*)
 - 5.1 Duodenum
 - 5.2 Jejunum
 - 5.3 Ileum
- 6 Large intestine (*intestinum crassum*)
 - 6.1 Caecum
 - 6.2 Colon
 - 6.3 Rectum
- 7 Liver (*hepar*)
- 8 Gallbladder (*vesica biliaris*)
- 9 Pancreas

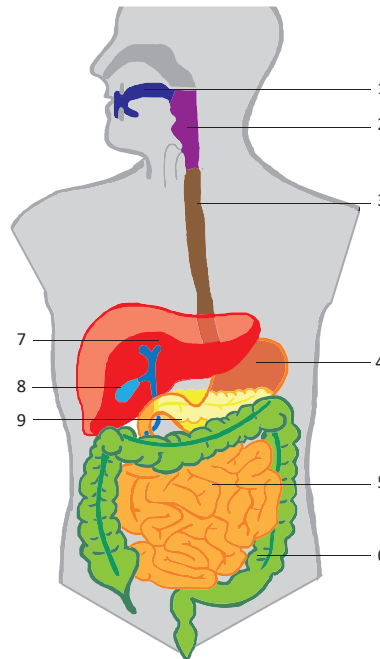
Position of individual parts

Organs in separate topographical spaces

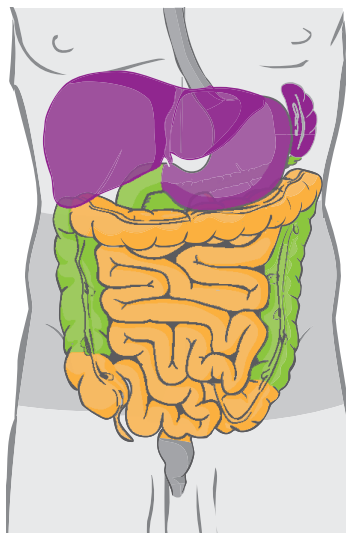
- The oral cavity is a topographical space of the skull
- The pharynx is located in the cervical visceral space
- The oesophagus extends from the cervical visceral space through the superior and posterior inferior mediastinum to the supramesocolic part of the abdominal cavity
- The aboral 2/3 of the rectum are located infraperitoneally

Position of organs in relation to the peritoneum

- **Supramesocolic part of abdominal cavity** (*pars supramesocolica cavitatis peritonealis*)
 - Oesophagus (abdominal part)
 - Stomach
 - Liver and gallbladder
 - Pancreas (upper portion of the head, body and tail)
 - Duodenum (superior part and upper portion of the descending part)
- **Inframesocolic part of abdominal cavity** (*pars inframesocolica cavitatis peritonealis*)
 - Jejunum and ileum
 - Caecum
 - Transverse colon
 - Sigmoid colon
 - Rectum (oral 1/3)
 - Pancreas (lower portion of the head, body and tail)
 - Duodenum (the inferior and ascending parts and the lower portion of the descending part)
- **Secondarily retroperitoneal organs**
 - Ascending colon
 - Descending colon
 - Pancreas (head and body)
 - Duodenum (except the ampulla)



Basic parts of the digestive system



Position of individual parts of the digestive system

The abbreviation **GIT** for Gastrointestinal Tract is often used for the digestive tract. The synonym alimentary tract is used less often.

For orientation in the digestive system, we use the terms oral (*oralis*), which means towards the mouth, and aboral (*aboralis*), which means away from the mouth (towards the anus).

Sphincters are made of thickened circular smooth muscle tissue. In some areas, such as the rectum, there are also skeletal muscle sphincters attached to the outer surface of the digestive canal.

The **enteric nervous system** is a special part of the autonomic nervous system that controls the digestive system. The submucous plexus of Meissner (*plexus submucosus*) and the myenteric plexus of Auerbach (*plexus myentericus*) work together. They are able to work independently of extrinsic signals, but both are influenced by the sympathetic and parasympathetic nervous systems.

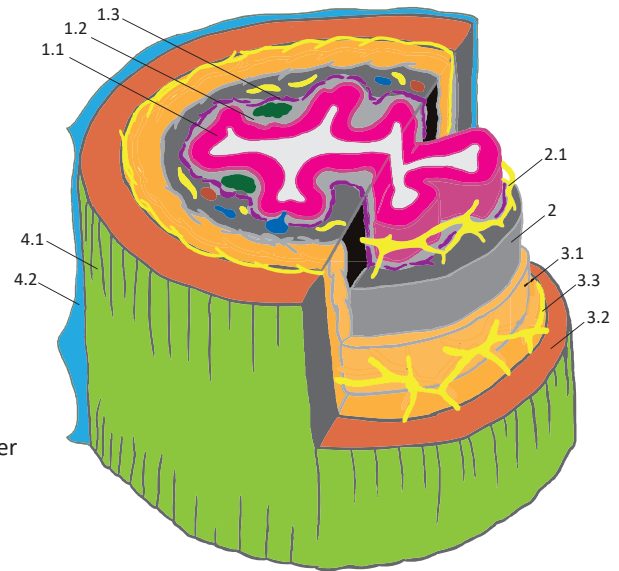
The **ileal diverticulum / Meckel's diverticulum** is a rudimentary relic of the omphaloenteric duct, which connects the lumen of embryonic intestine to the yolk sac and is connected at a distance of 0 to 100 cm orally to the ileocaecal transition. The diverticulum occurs only in 2 % of cases. In more than one half of cases, the diverticulum contains ectopic tissue (e.g. gastric, duodenal or pancreatic tissue), which can be a source of irritation or haemorrhage.

Glands connected to the digestive canal secrete their products through a system of ducts to the lumen of the digestive canal (exocrine glands) or into capillaries surrounding the cells of the glands (endocrine glands). Glands are located either in the wall of the digestive canal (intramural glands) or outside the wall (extramural glands).

The **ventral and dorsal mesogastrium** (*mesogastrium ventrale et dorsale*) are developmental peritoneal duplicatures (folds of peritoneum), which fix the primitive intestine (destined to become the stomach) to the anterior and posterior wall of the peritoneal cavity, respectively. During the embryonic period, the developing organs rotate around their duplicatures to reach their final positions.

Histology

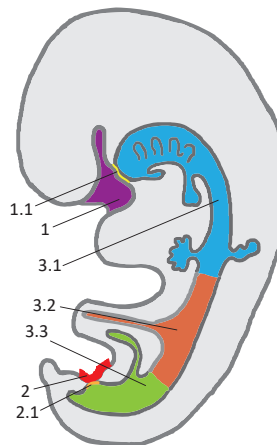
- 1 **Mucosa** (*tunica mucosa*) – also called mucous membrane
 - can either be smooth or contains papillae, villi or folds
 - 1.1 **Epithelium** – varies in character in different parts of the digestive canal
 - provides mechanical protection and resorbs and secretes substances
 - 1.2 **Lamina propria** (*lamina propria mucosae*) – a thin layer of loose connective tissue
 - contains lymphoid tissue (either diffuse or aggregated into nodules), mucous glands and nerve endings
 - 1.3 **Muscularis mucosae** (*lamina muscularis mucosae*)
 - a thin layer of smooth muscle that forms the border between the mucosa and submucosa
 - permits independent movement of the mucosa in relation to the other layers of the wall of the digestive canal
- 2 **Submucosa** (*tela submucosa*) – a layer of connective tissue located under the mucosa, which contains blood vessels, submucous glands, lymphoid tissue and a neural plexus
 - 2.1 **Submucous plexus of Meissner** (*plexus submucosus*) – an autonomic nervous plexus with ganglia
 - plays an important role in controlling the secretion of substances into the lumen and in regulating perfusion of the intestinal wall
 - works predominantly via local reflexes
- 3 **Muscular layer** (*tunica muscularis*) – the muscular layer of the oral cavity, pharynx, oral half of the oesophagus and aboral part of the rectum is composed of striated muscle
 - the muscular layer of all other parts of the digestive canal is composed of smooth muscle tissue
 - consists of two layers where it is made of smooth muscle tissue
 - 3.1 **Circular layer** (*stratum circulare*) – the inner layer
 - 3.2 **Longitudinal layer** (*stratum longitudinale*) – the outer layer
 - 3.3 **Myenteric plexus of Auerbach** (*plexus myentericus*)
 - an autonomic nervous plexus with ganglia located between the two layers of smooth muscle
 - is responsible for controlling intestinal peristalsis
- 4 **Outer surrounding connective tissue** (*tunica externa*) – the outermost layer
 - 4.1 **Adventitia** (*tunica adventitia*)
 - a layer of loose connective tissue covering and holding organs to the surroundings outside the peritoneal cavity
 - 4.2 **Serosa** (*tunica serosa*)
 - consists of subserous connective tissue covered by mesothelium
 - a glossy membrane covering the organs inside the peritoneal cavity
 - also called visceral peritoneum (*peritoneum viscerale*)



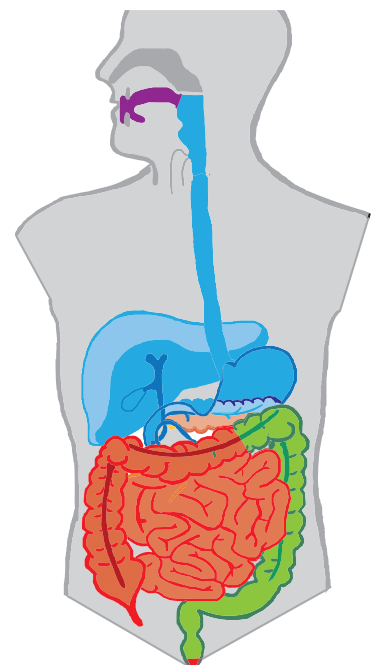
General structure of the wall of the digestive canal

Development of the digestive system

- 1 **Stomodeum** – the most anterior part of the developing digestive canal
 - is derived from the ectoderm and develops into the oral cavity
 - 1.1 **Oropharyngeal membrane** (*membrana oropharyngea*)
 - separates the stomodeum from the primitive intestine
- 2 **Proctodeum** – the most posterior part of the developing digestive canal
 - is derived from the ectoderm and develops into the aboral part of the anal canal
 - 2.1 **Cloacal membrane** (*membrana cloacalis*)
 - separates the proctodeum from the hindgut
- 3 **Primitive intestine** (*preenteron*) – is derived from the endoderm
 - 3.1 **Foregut** (*proenteron*) – extends from the oropharyngeal membrane to the liver bud
 - **develops into:** pharynx, oesophagus, stomach, duodenum (oral half), liver, gallbladder and pancreas
 - **supplied with blood by:** coeliac trunk
 - 3.2 **Midgut** (*mesenteron*) – opens ventrally into the yolk sac
 - **develops into:** duodenum (aboral half), jejunum, ileum, caecum, ascending colon, transverse colon (oral 2/3)
 - **supplied with blood by:** superior mesenteric artery
 - 3.3 **Hindgut** (*metenteron*) – extends from the end of the midgut to the cloacal membrane
 - **develops into:** transverse colon (aboral 1/3), descending colon, sigmoid colon, rectum (rectal ampulla and oral part of the anal canal)
 - **supplied with blood by:** inferior mesenteric artery



Embryonic digestive system



Adult digestive system

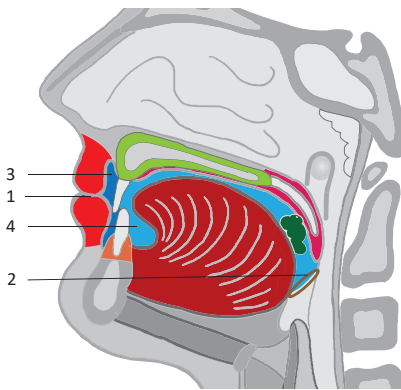
The **oral cavity** is bounded ventrally by the lips and laterally by the cheeks. The floor of the oral cavity is formed by the oral diaphragm and the roof by the palate. Dorsally, it is continuous through the isthmus of fauces with the pharynx. **The dental arches divide the oral cavity into the oral vestibule** ventral to the arches and **the oral cavity proper** dorsal to the arches.

Divisions and content

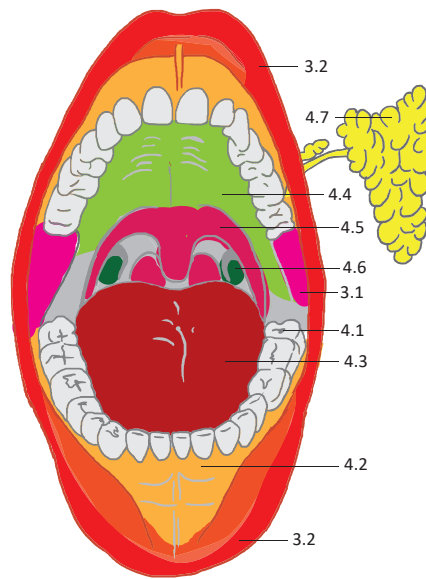
- 1 **Oral fissure** (*rima oris*) – the entrance to the oral cavity
- 2 **Isthmus of fauces / oropharyngeal isthmus** (*isthmus faucium*)
– marks the transition from the oral cavity to the oral part of the pharynx

The dental arches divide the oral cavity into:

- 3 **Oral vestibule** (*vestibulum oris*) – ventral to the dental arches
 - 3.1 **Cheek** (*bucca*) – contains the buccinator and the buccal fat pad
 - 3.2 **Lips** (*labia oris*) – surround the oral fissure
- 4 **Oral cavity proper** (*cavitas oris propria*) – dorsal to the dental arches
 - 4.1 **Teeth** (*dentes*)
 - 4.2 **Gum** (*gingiva*)
 - 4.3 **Tongue** (*lingua*)
 - 4.4 **Hard palate** (*palatum durum*)
 - 4.5 **Soft palate** (*palatum molle*)
 - 4.6 **Palatine tonsil** (*tonsilla palatina*)
 - 4.7 **Salivary glands** (*glandulae salivariae*)



The sagittal section of the oral cavity



Open oral cavity

The oral diaphragm (*diaphragma oris*) is a muscular plate under the tongue, which caudally encloses the oral cavity. It consists of the mylohyoid, geniohyoid and anterior belly of the digastric.

The oral vestibule and oral cavity proper communicate between the posterior edge of the third molar and the anterior edge of the coronoid process of the mandible. It is possible to use this gap to place a feeding tube when the mandible is immobilised in the treatment of a fracture. Small gaps between the teeth (*tremata*) are other places where the oral cavity proper and oral vestibule communicate.

We distinguish on the lips:

Cutaneous part: keratinised stratified squamous epithelium.

Transitional part (vermillion zone): The skin here does not contain any pigment and so the capillaries in the papillae of the dermis shine through, imparting a red-colour.

Vermillion border: the border between the cutaneous and transitional parts.

Mucous part: non-keratinised stratified squamous epithelium.

Cheek layers page 531.

Clinical notes

Facial clefts are caused by incomplete fusion of the lips, maxilla and hard palate.

Cheiloschisis – cleft of the lip

Gnathoschisis – cleft of the maxilla

Palatoschisis – cleft of the hard palate

Cheilognathopalatoschisis – combined cleft of all the above mentioned structures.

Anaemia is a deficiency of haemoglobin in the blood. The transitional zone of the lips is pale in this condition.

Cyanosis is caused by lack of oxygen in the blood, which increases the concentration of reduced haemoglobin. The transitional zone of the lips becomes violet in this condition.

The danger triangle of the face (*trigonum mortis*) is an area richly supplied with blood, located between the angles of the mouth and root of the nose. Venous blood from this area drains into the intracranial venous sinuses, thus there is a risk of a pathological process spreading to the intracranial space from this area.



2.1

Cheek – Bucca

The shape of the cheek is formed by the **buccinator**. The **buccal fat pad** (Bichat's fat pad) is located under the subcutaneous connective tissue and together with the fat in the subcutaneous connective tissue gives the cheek its definitive appearance.

Borders

The cheeks are bordered by the zygomatic arch **cranially**, the ramus of the mandible **caudally**, the angle of the mouth and the nasolabial sulcus **ventrally** and the posterior margin of the masseter **dorsally**.

Blood supply and innervation

Arterial supply: facial artery, transverse facial artery, buccal artery

Venous drainage: facial vein, transverse facial vein, deep facial vein

Lymphatic drainage: facial lymph nodes (on the outer surface of the buccinator), submandibular lymph nodes

Motor innervation: facial nerve (buccinator)

Sensory innervation: trigeminal nerve (skin)

The **lips** are two horizontal folds that surround the oral fissure and meet at the angle of the mouth. **The inner and outer sides of the lips are covered by different types epithelium**; there is a transitional zone between the two sides. The orbicularis oris forms the shape of the lips. The lower lip is attached to the paired mandibular processes. The upper lip is attached to the paired maxillary processes and the unpaired frontonasal process.

External structure

- 1 **Upper lip** (*labium superius*) – located between the oral fissure and lower margin of the nose, laterally bounded by the nasolabial sulcus
 - 1.1 **Frenulum of upper lip** (*frenulum labii superioris*) – attaches the upper lip to the gum from inside
 - 1.2 **Philtrum** – a sagittally oriented groove from the nasal septum to the upper lip
 - the margins of the philtrum are remnants of an adhesion of the three embryonic processes
 - 1.3 **Tubercle of upper lip** (*tuberculum labii superioris*) – tubercle on the lower end of the philtrum
- 2 **Lower lip** (*labium inferius*) – located between the oral fissure and the mentolabial sulcus
 - 2.1 **Frenulum of lower lip** (*frenulum labii inferioris*) – attaches the lower lip to the gum from inside
- 3 **Angle of mouth** (*angulus oris*) – a place of adhesion of the upper and lower lip
- 4 **Nasolabial sulcus** (*sulcus nasolabialis*) – a paired groove ongoing laterocaudally from the wing of the nose to the angle of the mouth
- 5 **Mentolabial sulcus** (*sulcus mentolabialis*) – an unpaired groove between the chin and lower lip

Blood supply

Arterial supply: superior and inferior labial artery
(form anastomosis in the superior and inferior labial arch)

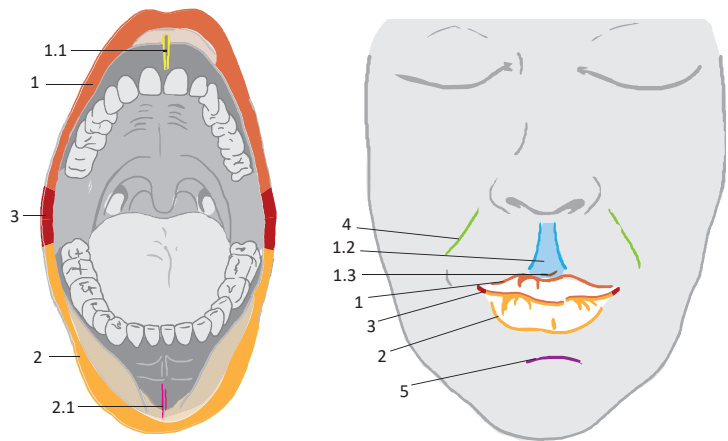
Venous drainage: facial vein

Lymphatic drainage:
submandibular and submental lymph nodes

Innervation

Motor innervation: facial nerve (facial muscles of the lips)

Sensory innervation: maxillary nerve (upper lip),
mandibular nerve (lower lip)



2.3

Gum – Gingiva

From the anatomical point of view, **the gum is a mucous membrane covering the alveolar processes of the maxilla and mandible**. From the clinical point of view the gum is divided into the **gum proper** and the **alveolar mucous membrane**. The gum proper is the mucosa located at the same level of the neck and crown of the tooth. It that does not contain any submucous connective tissue and is attached directly to the periosteum. It is thus known and as the mucoperisteum and is immobile. The alveolar mucosa covers the alveolar processes at the level of the root of the tooth. It contains submucosal connective tissue and is therefore mobile.

External structure

Alveolar gingiva / attached gum (*gingiva alveolaris*) – the gum attached to the alveolus (facing the oral cavity)

Marginal gingiva / free gingival margin (*gingiva marginalis*) – the gum attached to the tooth, facing the tooth and enclosing the entrance to the periodontal space

- 1 **Gingival margin** (*margo gingivalis*) – an elevated margin of the gum around the tooth
- 2 **Gingival sulcus** (*sulcus gingivalis*)
 - a groove between the margin of the gum and the tooth
 - 2.1 **Gingival recess** (*recessus gingivalis*) – a blind trunk of the gingival sulcus, which extends to the neck of the tooth
- 3 **Gingival papillae / interdental papillae** (*papillae gingivales*)
 - gingival projections between the necks of the teeth

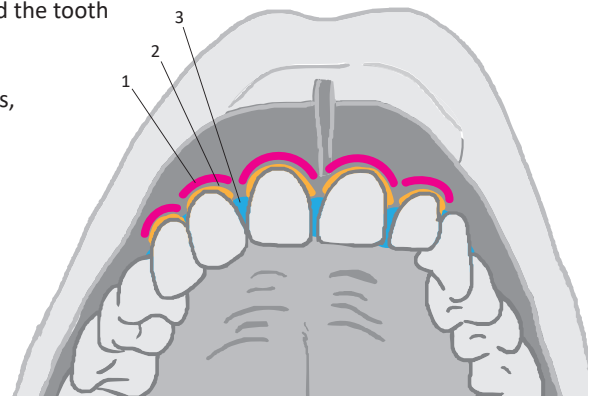
Blood supply and innervation

Arterial supply: superior alveolar arteries, inferior alveolar arteries

Venous drainage: pterygoid plexus, facial vein, lingual vein

Lymphatic drainage: submandibular lymph nodes

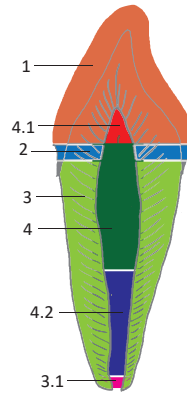
Mixed innervation: superior dental plexus (maxillary nerve),
inferior dental plexus (mandibular nerve)



The **teeth** are held in the dental alveoli by a specific type of joint called a **gomphosis**. They are arranged in two arches: **the maxillary and mandibular arcades**. Eruption of **deciduous teeth** begins in the 6th month of life and ends in the 24th month. There are 20 deciduous teeth in total. Eruption of **permanent teeth** begins in the 6th year and is usually complete by the 30th year. There are 32 permanent teeth in total. The main function of the teeth is **mechanical digestion**. However they also take part in voice production and maintenance of the proper position of the temporomandibular joint.

External structures

- 1 **Crown of tooth** (*corona dentis*)
 - the visible part of the tooth, covered by enamel
- 2 **Neck of tooth** (*cervix dentis*)
 - the transition between the crown and root, covered by the gum
- 3 **Root of tooth** (*radix dentis*) – is placed in the dental alveolus of the jaw
 - 3.1 **Root apex** (*apex radialis dentis*) – contains an opening (*foramen apicis radialis dentis*) where vessels and nerves enter the tooth
- 4 **Pulp cavity** (*cavitas dentis*) – contains the dental pulp (*pulpa dentis*), which is connective tissue that has a rich neurovascular supply
 - 4.1 **Pulp cavity of crown** (*cavitas coronae*)
 - a widening of the pulp cavity within the crown of the tooth
 - 4.2 **Root canal** (*canalis radialis dentis*)
 - a narrowing of the pulp cavity within the root apex



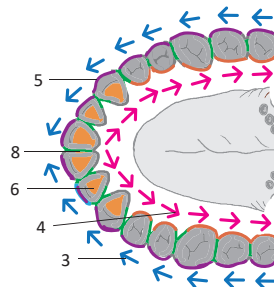
Directions and surfaces on the dental crown

Directions

- 1 **Cervical** – towards the neck of the tooth
- 2 **Apical** – towards the root of the tooth
- 3 **Mesial** – towards the anterior midline of the dental arcade
- 4 **Distal** – towards the posterior teeth

Surfaces

- 5 **Vestibular surface** (*facies vestibularis*) – faces the oral vestibule
 - 5.1 **Labial surface** (*facies labialis*) – faces the lips (the anterior teeth)
 - 5.2 **Vestibular surface** (*facies vestibularis*) – faces the cheeks (the posterior teeth)
- 6 **Lingual surface** (*facies lingualis*) – faces the tongue (the lower teeth)
- 7 **Palatal surface** (*facies palatinalis*) – faces the palate (the upper teeth)
- 8 **Approximal surface** (*facies approximalis*) – faces the neighboring tooth
- 9 **Occlusal surface** (*facies occlusalis*)
 - the chewing surface at the top of the crown



Histology

- 10 **Dentine** (*dentinum*) – makes up the basic substance of the tooth
- 11 **Enamel** (*enamelum*) – covers the dentin of the crown
- 12 **Cement** (*cementum*) – covers the neck and root of the tooth

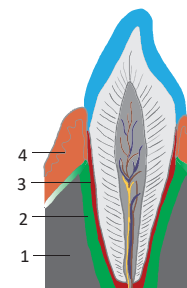
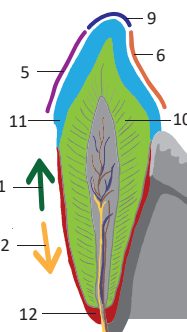
Fixation of the teeth in the jaw

Parodontium – a complex of structures that supports the teeth

- 1 **Dental alveolus** (*alveolus dentalis*) – a fossa for the tooth that is made from compact bone
- 2 **Periodontium** – the dentoalveolar connection, continues cranially into the connective tissue of the gum
 - 2.1 **Gomphosis** (a type of syndesmosis) – a fibrous connection between the root of the tooth and the periosteum on the inner surface of the dental alveolus
- 3 **Cement** – a layer of fibrillar bone on the surface covering the root and neck of the tooth
- 4 **Gingiva / gum** – supports the teeth at the neck of the tooth

Dentogingival junction (*junctio dentogingivalis*)

- the gingiva covers the periodontium around the neck of the tooth
- a special junction between the gingival epithelium and the enamel and cement
- there is an infiltration of lymphocytes and plasmocytes in the connective tissue beneath the dentogingival junction, which forms an immunological barrier



Odus (genitive: *odontos*) is the greek term for tooth.

Dental articulation (gliding occlusion) is term for the mutual contact of the occlusal surfaces of the upper and lower teeth.

Neutroclusion is a form of occlusion characterised by the upper teeth being located anterior to the lower teeth.

Crossbite is a form of occlusion when part of the lower dental arcade overlaps with the upper arcade.

The old system of the dental formula was based on the first letter of the Latin name of each tooth and its location in the dental arcade:

Deciduous teeth – i1, i2, m1, c, m2
Permanent teeth – M1, I1, I2, P1, C, P2, M2, M3.

A mnemonic for the sequence of permanent teeth eruption based on the old system of the dental formula is: “Mama Is In Pain, Papa Can Make Medicine”.

The first molar (M1) is under the most pressure. **The third molars (M3)** erupt in adulthood and are also called wisdom teeth.

According to the knowledge of tooth eruption, it is possible to estimate the age of a child with a discrepancy of about half a year.

Contact surface of the tooth (*facies contactus*) is an obsolete term for the approximal surface of a tooth.

Clinical notes

The periodontal gap is a space richly innervated and supplied with blood, located between the tooth and alveolus where the periodontium is located.

The periapical space is a widening of the periodontal gap around the root apex. Inflammatory of teeth is most likely to occur here .

Occlusal abrasion is abrasion of the teeth caused by wearing out during life.

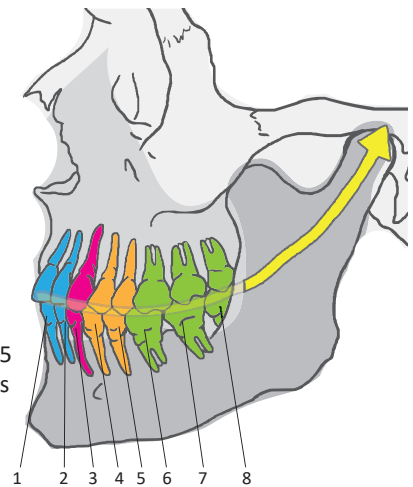
Gingival recession (receding gums) is a loss of gum tissue caudal to the crown of the tooth. It exposes the neck of the tooth.

A reduction of the alveolar processes is a decrease in height of the alveolar bones.

An orthopantomograph (OPG) is an X-ray image that clearly shows the facial part of the skull with the teeth in both dental arcades.

Dental arcades and types of teeth

- **Maxillary/upper dental arcade** (*arcus dentalis maxillaris/superior*) – is elliptical-shaped – is larger than the lower dental arcade, which it ventrally and laterally overlaps
- **Mandibular/lower dental arcade** (*arcus dentalis mandibularis/inferior*) – is parabolic-shaped
- **Articulation curve** – runs from the incisor through the molars to the temporomandibular joint – is produced by the masticatory muscles, which create a force directed towards the first molar
- 1, 2 **Incisor teeth** (*dentes incisivi*) (**I**) – contain one root and usually one root canal – there are two incisors in each quadrant, which are referred to by the numbers 1 and 2
- 3 **Canine teeth** (*dentes canini*) (**C**) – contain one root and usually one root canal – there is one canine in each quadrant, which is referred to by the number 3
- 4, 5 **Premolar teeth** (*dentes premolares*) (**P**) – contain one root and usually one root canal but the upper first premolar has two roots and two canals – there are two premolars in each quadrant, which are referred to by the numbers 4 and 5
- 6, 7, 8 **Molar teeth** (*dentes molares*) (**M**) – upper molars usually have three roots and four canals – lower molars usually have two roots and three canals – there are three molars in each quadrant, which are referred to by the numbers 6, 7, 8



Dental formula

The teeth are divided into four quadrants seen in an open mouth.

- **Deciduous teeth** (*dentes decidui*) – 20 teeth divided into four quadrants, each containing 5 teeth (51–55, 61–65, 71–75, 81–85)
- **Permanent teeth** (*dentes permanentes*) – 32 teeth divided into four quadrants, each containing eight teeth (11–18, 21–28, 31–38, 41–48)

55	54	53	52	51	61	62	63	64	65
85	84	83	82	81	71	72	73	74	75

Dental formula of the deciduous teeth

Quadrants are described in a clockwise direction from the right upper quadrant:

right upper quadrant → left upper quadrant

right lower quadrant ← left lower quadrant

18	17	16	15	14	13	12	11	21	22	23	24	25	26	27	28
48	47	46	45	44	43	42	41	31	32	33	34	35	36	37	38

Dental formula of the permanent teeth

Tooth eruption

Eruption of the deciduous teeth

- 6th–8th month – **incisors** (i1), upper first, then lower
- 7th–12th month – **incisors** (i2), upper first, then lower
- 12th–16th month – **molars** (m1), lower first, then upper
- 15th–20th month – **canines** (c), lower first, then upper
- 20th–30th month – **molars** (m2), upper and lower together

Eruption of the permanent teeth

- 6th–8th year – **molars** (M1), usually lower first
- 6th–7th year – **incisors** (I1), usually lower first
- 7th–9th year – **incisors** (I2), usually lower first
- 9th–11th year – **premolars** (P1)
- 9th–14th year – **canines** (C)
- 11th–14th year – **premolars** (P2)
- 10th–15th year – **molars** (M2)
- 17th–30th year – **molars** (M3)



Simplified panoramic dental X-ray image (OPG) of an 8.5-years-old boy

Blood supply and innervation

Artery supply:

- **the maxillary (upper) dental arcade:** dental branches (*rr. dentales*) from the posterior superior alveolar artery (maxillary artery) and from the anterior superior alveolar arteries (infraorbital artery from the maxillary artery)
- **the mandibular (lower) dental arcade:** dental branches (*rr. dentales*) from the inferior alveolar artery (maxillary artery)

Venous drainage: to the pterygoid plexus and further to the maxillary veins (to the retromandibular vein)

Lymphatic drainage: submandibular lymph nodes, deep cervical lymph nodes

Somatosensory:

- **the maxillary (upper) dental arcade:** dental branches (*rr. dentales*) from the superior dental plexus (anterior, middle and posterior superior alveolar branches from the infraorbital nerve from the maxillary nerve)
- **the mandibular (lower) arcade:** dental branches (*rr. dentales*) from the inferior dental plexus (inferior alveolar nerve from the mandibular nerve)

The **tongue** is a muscular organ, covered by mucosa and located in the oral cavity. The muscles of the tongue are divided into two groups: the intrinsic and extrinsic muscles. The **intrinsic muscles** are entirely located within the tongue itself and function to **change the shape of the tongue**. The **extrinsic muscles** attach to the tongue from the surrounding structures and function to **change the position of the tongue**. The tongue helps to **mechanically break down food**, participates in both **articulation** and **phonation** and, by means of its taste buds, in **taste perception**.

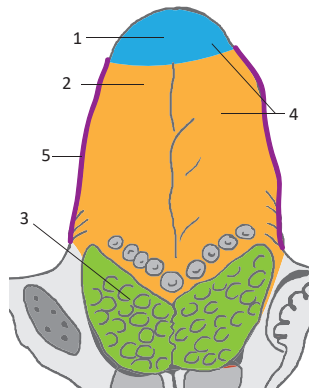
Parts and surfaces of the tongue

Parts

- 1 **Apex/tip of tongue** (*apex linguae*)
– the most ventral part of the tongue
- 2 **Body of tongue** (*corpus linguae*)
– the ventral 2/3 of the tongue
- 3 **Root of tongue** (*radix linguae*)
– the dorsal 1/3 of the tongue

Surfaces

- 4 **Dorsum of tongue** (*dorsum linguae*)
– faces the roof of the oral cavity
- 5 **Margin of tongue** (*margo linguae*) – faces the teeth
- 6 **Inferior surface of tongue** (*facies inferior linguae*)
– faces the oral diaphragm



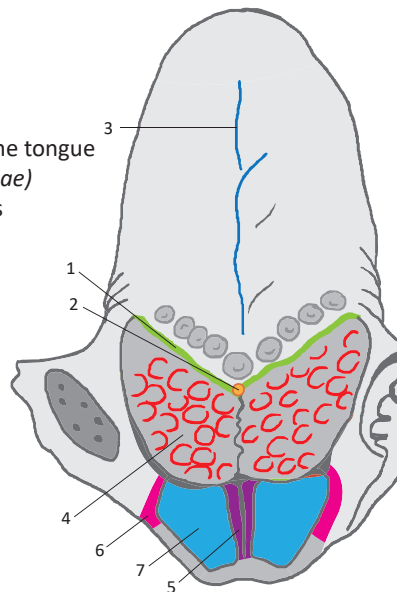
Structures on the dorsum and root of the tongue

Structures on the dorsum of the tongue

- 1 **Terminal sulcus of tongue** (*sulcus terminalis linguae*)
– a “V” shaped groove with a dorsally pointing vertex
– bordered ventrally by the 8–12 vallate papillae
– forms the border between the body and the root of the tongue
- 2 **Foramen caecum of the tongue** (*foramen caecum linguae*)
– a small depression at the vertex of the terminal sulcus
- 3 **Midline groove of tongue / median sulcus of tongue** (*sulcus medianus linguae*)
– a groove in the median plane of the tongue
– the superficial projection of the lingual septum

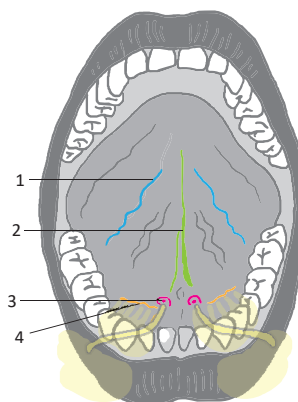
Structures on the root of the tongue

- 4 **Lingual tonsil** (*tonsilla lingualis*)
– diffuse lymphoid tissue located in the submucosal connective tissue that forms bumps on the surface of the root of the tongue
- 5 **Median glosso-epiglottic fold** (*plica glossoepiglottica mediana*)
– a sagittal mucosal fold extending dorsally from the root of the tongue to the epiglottis
- 6 **Lateral glosso-epiglottic folds** (*plicae glossoepiglotticae laterales*)
– two mucosal folds located lateral to the epiglottic valleculae
- 7 **Epiglottic valleculae** (*valleculae epiglotticae*)
– two depressions between the lateral glosso-epiglottic folds and median glosso-epiglottic fold



Structures on the inferior surface of the tongue

- 1 **Fimbriated fold** (*plica fimbriata*) – a paired dorsolateral fold
- 2 **Frenulum of tongue** (*frenulum linguae*)
– a sagittal mucosal fold in the median plane connecting the inferior surface of the tongue with the floor of the oral cavity
- 3 **Sublingual caruncle** (*caruncula sublingualis*)
– a paired tubercle lateral to the frenulum of the tongue where the submandibular and major sublingual ducts open
- 4 **Sublingual fold** (*plica sublingualis*) – a paired dorsolateral fold where the 8–30 minor sublingual ducts open



Glossa is the Greek term for tongue.

The **mucosa of the tongue** is covered by stratified squamous epithelium, which keratinises in some places and forms projections of various shapes called lingual papillae.

The **vallate papillae** (*papillae vallatae*) are the largest lingual papillae and are located ventral to the terminal sulcus of the tongue. Taste buds are located in the epithelium around the vallate papillae (as well as in various other parts of the oral cavity). Other lingual papillae include the filiform, fungiform and foliate papillae. The foliate papillae are rudimentary leaf-like papillae on the margin of the tongue.

The **foramen caecum of the tongue** is a remnant of the thyroglossal duct. The thyroglossal duct is a canal formed by the descent of the thyroid during development.

The **lingual septum** fades out basally and changes into a sagittally oriented space filled with loose connective tissue called the middle space of the tongue (*spatium medianum linguae*), which extends caudally to the mylohyoids, ventrally to the mental spines of the mandible and dorsally to the hyoid bone.

The palatoglossus

- functionally belongs to the muscles of the tongue
- developmentally belongs to the muscles of the soft palate
- innervation: vagus nerve

The **palolingual canal** is a slit between the hyoglossus and genioglossus, where the lingual artery and vein pass.

Clinical notes

When performing first aid, the tongue must be prevented from falling backwards and blocking the airways. This is done by moving the mandible forward (as the tongue is attached to the mandible by the genioglossus). The same effect can be achieved by bending the head backwards, which causes the root of tongue to move out of the laryngeal part of the pharynx.

The **manoeuvre of Esmarch** is performed by simultaneously bending the head backwards, moving the mandible forwards and opening the mouth.

Ankyloglossia (tongue-tie) is a congenital abnormality of the tongue characterised by a short and tough frenulum. This condition impedes the ability to suckle and later to speak.

Muscles of the tongue (*musculi linguae*)

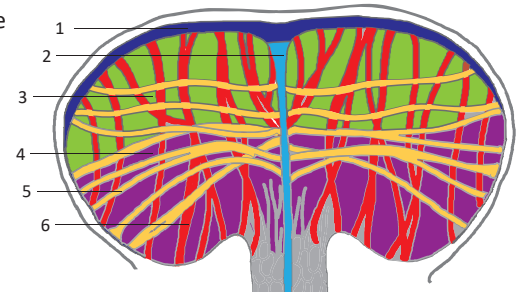
All muscles of the tongue (intrinsic and extrinsic tongue muscles) attach to the lingual aponeurosis or the lingual septum.

- 1 **Lingual aponeurosis** (*aponeurosis linguae*) – a fibrous disc beneath the mucosa of the dorsum of the tongue
- 2 **Lingual septum** (*septum linguae*) – an incomplete sagittal fibrous partition perpendicular to the lingual aponeurosis

Intrinsic (intraglossal) muscles

The intrinsic muscles of the tongue change the tongue's shape. The muscle fibres are interlaced and do not contain any fasciae.

- 3 **Superior longitudinal muscle** (*musculus longitudinalis superior*) – shortens the length of the tongue and arches the dorsum of tongue
 - runs sagittally beneath the lingual aponeurosis to which it is connected
- 4 **Inferior longitudinal muscle** (*musculus longitudinalis inferior*) – shortens the tongue
 - runs sagittally above the genioglossus
 - its fibres interlace with the genioglossus and hyoglossus
- 5 **Transverse muscle** (*musculus transversus*) – narrows and extends the tongue
 - runs across the tongue from the sides through the lingual septum
- 6 **Vertical muscle** (*musculus verticalis*) – flattens, broadens and extends the tongue
 - runs between the dorsal and inferior surfaces of the tongue

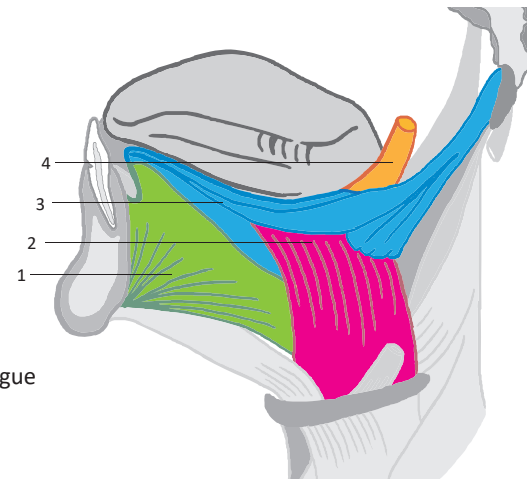


Frontal section of the tongue

Extrinsic (extraglossal) muscles

The extrinsic muscles of the tongue originate on the surrounding structures and insert on the fibrous lingual apparatus. They change the position of the tongue.

- 1 **Genioglossus** (*musculus genioglossus*)
 - Origin:** superior mental spine
 - Insertion:** lingual aponeurosis
 - Function:** protrusion and depression of the tongue and protects the tongue from falling backwards
- 2 **Hyoglossus** (*musculus hyoglossus*)
 - Origin:** the greater horn of the hyoid bone
 - Insertion:** lateral to the genioglossus
 - Action:** depression and retraction of the tongue
- 3 **Styloglossus** (*musculus styloglossus*)
 - Origin:** styloid process and stylomandibular ligament
 - Insertion:** the tip and sides of the tongue
 - Function:** retraction of the tongue and elevation of the body and apex of the tongue
- 4 **Palatoglossus** (*musculus palatoglossus*)
 - Origin:** the lateral margins of the palatine aponeurosis
 - Course:** within the palatoglossal arch (forms the anterior pillar of the fauces)
 - Insertion:** the root of the tongue
 - Function:** elevation of the root of the tongue and depression of the soft palate, thus the final action is to narrow the opening of the pharynx



Sagittal section of the mandible and extrinsic muscles of the tongue, lateral view

Blood supply

Arterial supply: lingual artery (a branch of the external carotid artery)

Venous drainage: lingual vein, vena comitans of the hypoglossal nerve (to the internal jugular vein)

Lymphatic drainage: from the apex – submental lymph nodes, from the margins – submandibular lymph nodes, from the body and root – deep cervical lymph nodes, due to the incomplete septum of the tongue also to the contralateral nodes

Innervation

Motor innervation: hypoglossal nerve – all muscles except the palatoglossus (vagus nerve)

Somatosensory innervation:

lingual nerve (a branch of the mandibular nerve), glossopharyngeal nerve, vagus nerve

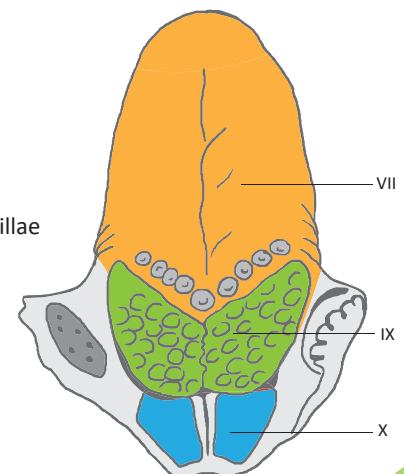
Sensory innervation – taste:

- VII **Chorda tympani** – provides taste on the apex and body of the tongue except the vallate papillae
 - a branch of the facial nerve, from the tongue fibers travel via the lingual nerve
- IX **Glossopharyngeal nerve** – provides taste on the root of the tongue and vallate papillae
- X **Vagus nerve** – provides taste on the epiglottic valleculae

Parasympathetic innervation (of the salivary glands):

- VII **Chorda tympani** – innervates the glands within the apex and body of the tongue
- IX **Glossopharyngeal nerve** – innervates the glands within the root of the tongue
- X **Vagus nerve** – innervates the glands within the epiglottic valleculae

Sympathetic innervation: via the periarterial plexus surrounding the lingual artery



The **palate** is a horizontal partition separating the oral cavity from the nasal cavity. The ventral part is formed by the **bony hard palate**. The dorsal part is called the **soft palate** and formed by a fibromuscular plate. The palate is used during swallowing, phonation and articulation.

2.6.1

Hard palate – Palatum durum

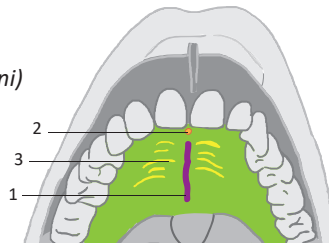
The **hard palate** is a horizontal bony partition covered by mucosa and separating the oral cavity from the nasal cavity.

Components

- 1 **Incisive bone** (*premaxilla, os incisivum*) - unpaired
- 2 **Palatine process of maxilla** (*processus palatinus maxillae*)
- 3 **Horizontal plate of palatine bone** (*lamina horizontalis ossis palatini*)

Structures

- 1 **Palatine raphe** (*raphe palati*)
– a mucosal ridge running in the median plane
- 2 **Incisive papilla** (*papilla incisiva*) – elevated mucosa above the opening of the incisive canal on the ventral part of the palatine raphe
- 3 **Transverse palatine folds** (*palatine rugae, plicae palatinae transversae, rugae palatinae*)
– low transverse mucous folds on the oral side of the palate



2.7

Palatine tonsil – Tonsilla palatina

The palatine tonsil is an **oval-shaped lymphoid organ** covered by mucosa and located in the tonsillar fossa/sinus (*fossa tonsillaris*) **between the palatoglossal and palatopharyngeal arches**. It is a part of the immunological barrier that prevents infection from spreading from the oral and nasal cavity. The surface of the tonsil is decorated by the tonsillar pits and crypts.

Syntopy

Medially: isthmus of fauces

Laterally (in layers):

1. Tonsillar capsule (*capsula tonsillae*) – a fibrous capsule that borders the lateral aspect of the tonsil
– emits septa into the surrounding tissue
2. Peritonsillar space – a space filled with loose connective tissue
– is continuous with the surrounding submucosal connective tissue
– contains the venous tonsillar plexus and the external palatine vein
3. Superior constrictor of the pharynx (*musculus constrictor pharyngis superior*)
4. Parapharyngeal space (*spatium parapharyngeum*)

Cranially (structures):

- 1 Supratonsillar fossa (*fossa supratonsillaris*)
- 2 Semilunar fold (*plica semilunaris*)
- 3 Triangular fold (*plica triangularis*)

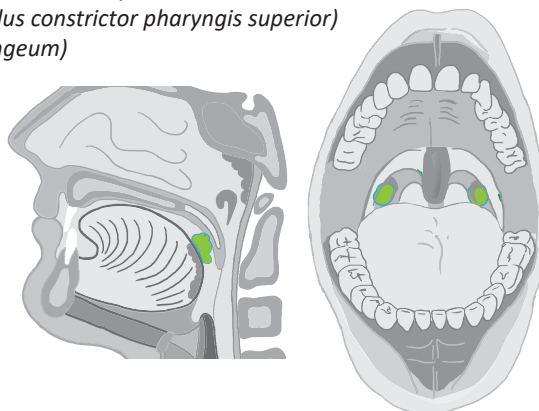
Blood supply and innervation

Arterial supply: ascending palatine artery, lesser palatine arteries, lingual artery, ascending pharyngeal artery

Venous drainage: external palatine vein

Lymphatic drainage: into the deep cervical lymph nodes but mostly to the tonsillar lymph node (node of Wood) located behind the angle of the mandible

Sensory innervation: maxillary nerve, glossopharyngeal nerve



Velum is an obsolete term for the soft palate.

A mnemonic for the palatine muscles:

2 muscles are attached directly to the palate:

- tensor veli palatini and levator veli palatini
- complicated position, course and innervation

1 muscle is simple:

- musculus uvulae

2 muscles are in arches:

- palatoglossus and palatopharyngeus
- both run within the homonymous arches
- developmentally are classified as muscles of the pharynx
- functionally are classified as muscles of the tongue and pharynx
- both are elevators
- both are innervated by the vagus nerve

Clinical notes

During acute tonsillitis dead bacteria and denuded epithelial cells accumulate in the pits and crypts of the tonsil and a white exudate emerges on the surface. Swollen tonsils can narrow or even completely close the lumen of the pharynx, impairing swallowing or even causing asphyxiation. Inflammation of the palatine tonsils is also called angina tonsillaris.

Tonsillectomy is a surgical procedure to remove the tonsils from the peritonsillar space. In the loose connective tissue of this space, inflammation can arise and spread to the surrounding tissue. This condition can deteriorate to the point of a life-threatening cervical phlegmon and spread to the mediastinum.

A peritonsillar abscess is a bordered inflammatory focus in the peritonsillar space. It is a frequent complication of a bacterial inflammation of the palatine tonsil.

A paratonsillar abscess is a type peritonsillar abscesses. It is located in the peritonsillar space lateral to the tonsil and can extend into the parapharyngeal space.

The internal carotid artery runs close to the palatine tonsil in the parapharyngeal space. It can be damaged during a careless tonsillectomy as it runs about 25 mm dorsal to the tonsil. Nevertheless, the most common structure prone to bleed during and after tonsillectomy is the paratonsillar vein (from the external palatine vein).

The soft palate is a fibromuscular plate that consists of the the palatine aponeurosis and the muscles that insert and originate on it. Two palatine arches extend laterocaudally from the soft palate: the palatoglossal arch and palatopharyngeal arch. They are formed by two homonymous muscles. The uvula protrudes caudally from the posterior part of the soft palate. The soft palate **participates in swallowing and prevents food from passing into the nasopharynx.**

Structures

- 1 **Uvula** (*uvula palatina*) – a caudally protruding part of the soft palate
- 2 **Palatoglossal arch / anterior pillar of fauces** (*arcus palatoglossus*)
– an arched elevation between the palate and tongue, formed by the palatoglossus
- 3 **Palatopharyngeal arch / posterior pillar of fauces** (*arcus palatopharyngeus*)
– an arched elevation between the palate and pharynx, formed by the palatopharyngeus

Muscles

- 1 **Tensor veli palatini** (*musculus tensor veli palatini*) – tenses the soft palate
Origin: spine of sphenoid bone, scaphoid fossa of sphenoid bone, lateral side of the cartilaginous part of the auditory tube
Course: descends laterally from the medial lamina of the pterygoid process of sphenoid bone, its inserting tendon turns medially around the pterygoid hamulus
Insertion: palatine aponeurosis
Innervation: mandibular nerve (branch of the trigeminal nerve)
Function: tenses the soft palate and narrows the auditory tube by pressing its tendon on the wall of the auditory tube
- 2 **Levator veli palatini** (*musculus levator veli palatini*) – the levator of the soft palate
Origin: apex of petrous part of temporal bone, medial wall of the cartilaginous part of auditory tube
Course: descends medially from the medial lamina of the pterygoid process of the sphenoid bone
Insertion: palatine aponeurosis
Innervation: vagus nerve
Function: elevates the soft palate
- 3 **Musculus uvulae**
Origin: posterior nasal spine of palatine bone and palatine aponeurosis
Insertion: connective tissue of the uvula, it protrudes into the pharynx (the fauces)
Innervation: vagus nerve
Action: shortens and elevates the uvula
- 4 **Palatoglossus** (*musculus palatoglossus*)
Origin: the lateral margins of the palatine aponeurosis
Course: within the palatoglossal arch
Insertion: the root of the tongue
Innervation: vagus nerve
Function: elevation of the root of the tongue and depression of the soft palate, thus the final action is narrowing of the opening of the pharynx
- 5 **Palatopharyngeus** (*musculus palatopharyngeus*)
Origin: palatine aponeurosis and medial lamina of pterygoid process of sphenoid bone, pterygoid hamulus and the inferior part of the auditory tube
Course: within the palatopharyngeal arch
Insertion: posterior margin of the thyroid cartilage and the wall of the pharynx
Innervation: vagus nerve
Action: elevation of the pharynx and depression of the soft palate, thus the final action is narrowing of the opening of the pharynx (the fauces)

Blood supply and innervation

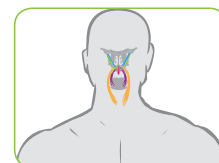
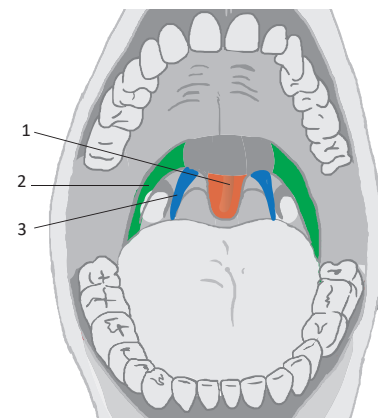
Arterial supply: descending palatine artery, ascending palatine artery, ascending pharyngeal artery, lingual artery

Venous drainage: pterygoid plexus, pharyngeal veins

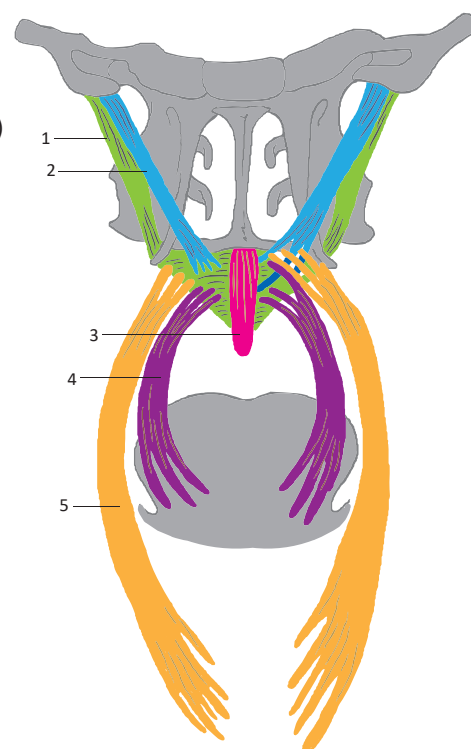
Motor innervation: vagus nerve, except for the tensor veli palatini muscle, which is innervated by the mandibular nerve

Somatosensory and parasympathetic innervation: greater palatine nerve, lesser palatine nerves, nasopalatine nerve (maxillary nerve – somatosensory, facial nerve – parasympathetic and taste)

Sympathetic innervation: pharyngeal plexus (glossopharyngeal nerve, vagus nerve)



Position of the muscles of soft palate and fauces, posterior view



Posterior view of the muscles of the soft palate and fauces

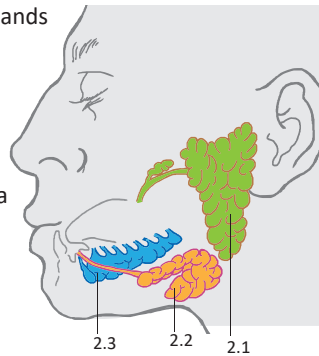
The **salivary glands** are connected by their ducts to the oral cavity. They produce **saliva**, which has a serous and a mucous component. **The serous component** contains enzymes, which participate in the initial breakdown of food. **The mucous component** lubricates the ingested food and prepares it for swallowing. Saliva secretion is regulated by reflexes and **controlled directly by the autonomic nervous system**. The autonomic nervous system processes stimuli that lead to salivation, such as: smell, taste, visual stimuli and somatosensory stimuli, made by contact between food and the mucosa of the oral cavity.

Types of glands according to the type of saliva they produce

- 1 **Serous glands:** the parotid gland and the gustatory glands (von Ebner's glands)
- 2 **Mucous glands:** glands of the root of the tongue (Weber's glands), the posterior part of the sublingual gland and the palatine glands
- 3 **Seromucous glands:** the submandibular gland, the anterior part of the sublingual gland, labial glands, buccal glands, molar glands and the anterior lingual glands

Types of glands according to their size

- 1 **Minor salivary glands** (*glandulae salivariae minores*) – the lesser glands
 - scattered throughout the oral cavity, embedded in mucosa, submucosa and sometimes even in the muscular layer
 - produce saliva continuously without neuronal stimuli
 - 1.1 **Labial glands** (*glandulae labiales*) – glands of the lip
 - 1.2 **Buccal glands** (*glandulae buccales*) – glands of the cheek
 - 1.3 **Molar glands** (*glandulae molares*) – glands of the molar area
 - 1.4 **Palatine glands** (*glandulae palatinae*) – glands of the palate
 - 1.5 **Lingual glands** (*glandulae linguales*) – glands of the tongue
- 2 **Major salivary glands** (*glandulae salivariae majores*)
 - open via ducts into the oral cavity
 - produce saliva in response to neuronal stimuli
 - 2.1 **Parotid gland** (*glandula parotidea*) – is located around the ramus of the mandible
 - 2.2 **Submandibular gland** (*glandula submandibularis*) – is located beneath the mandible
 - 2.3 **Sublingual gland** (*glandula sublingualis*)
 - is located beneath the tongue and above the mylohyoid



Sublingual gland (*glandula sublingualis*)

The sublingual gland is located in the sublingual space. The anterior part of the gland is seromucous. The posterior part is purely mucous.

Syntopy

Caudally: in contact with the upper surface of the mylohyoid

Cranially: forms a prominence in the oral cavity called the sublingual fold

Laterally: in contact with the sublingual fovea of the mandible

Medially: in contact with the hyoglossus, geniohyoid, submandibular duct and lingual nerve

Ducts

- 1 **Major sublingual duct / duct of Bartholin** (*ductus sublingualis major*)
 - leaves the anterior seromucous part of the gland
 - its opening is located on the sublingual caruncle with the opening of the submandibular duct
- 2 **Minor sublingual ducts** (*ductus sublinguales minores*)
 - 8–20 ducts that leave from the posterior part of the gland
 - are located on each side of the gland and open on the sublingual fold

Blood supply and innervation:

Arterial supply: facial artery, lingual artery

Venous drainage: facial vein, lingual vein

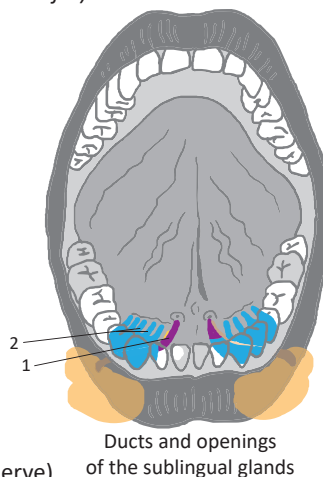
Lymphatic drainage: submandibular lymph nodes

Parasympathetic innervation:

glandular branches from the submandibular ganglion (fibres from the facial nerve via the chorda tympani)

Sympathetic innervation: periarterial plexus

Viscerosensory innervation: lingual nerve (from the mandibular nerve)



Ducts and openings of the sublingual glands

Von Ebner's glands (gustatory glands) are serous glands that wash out the grooves around the vallate papillae where taste buds are located.

Weber's glands (glands of the root of the tongue) are mucous glands. They protect the root of the tongue.

The retromandibular fossa is located between the lateral margin of the ramus of the mandible, external acoustic meatus, posterior belly of the digastric, styloid process, styloid septum and pterygoids.

Relation of the parotid gland to nerves in its proximity:

The facial nerve enters the gland after passing through the stylomastoid foramen. Within the gland it ramifies forming the intraparotid plexus, which divides the gland into superficial and deep parts.

The auriculotemporal nerve passes behind the parotid gland to reach the temporal region.

Division of the parotid gland by the intraparotid plexus (branches of the facial nerve) into the superficial and deep part does not match exactly the morphological division into the pars superficialis and pars profunda.

Parasympathetic fibres that supply the parotid gland originate in the inferior salivary nucleus. They travel in the tympanic branch of the glossopharyngeal nerve to the middle ear before leaving the skull as the lesser petrosal nerve. They synapse in the otic ganglion and reach the parotid gland through the parotid branches of the auriculotemporal nerve.

The salivary glands produce 1 to 1.5 liters of saliva daily. Saliva is important for hygiene of the oral cavity, lubricating food and in preventing the formation of dental calculus.

Clinical notes

Sialolithiasis refers to the presence of stones in the salivary ducts. This occurs most frequently in the submandibular duct due to its upward course.

Sialoadenitis is inflammation of the salivary gland. This can occur as a consequence of sialolithiasis.

Epidemic parotitis (mumps) is an acute viral inflammation of the parotid gland. Its clinical symptoms are swelling and pain. The disease can be accompanied by inflammation of other glands such as the pancreas and gonads. Infection of the male gonads may result in sterility.

Submandibular gland (*glandula submandibularis*)

The submandibular gland is a seromucous gland in the submandibular space. It secretes predominantly serous saliva.

Syntopy

Dorsally: is separated from the parotid gland by the stylomandibular ligament

Laterally: faces the submandibular fovea of the mandible

Turns around the posterior margin of the mylohyoid and follows its duct to the sublingual space.

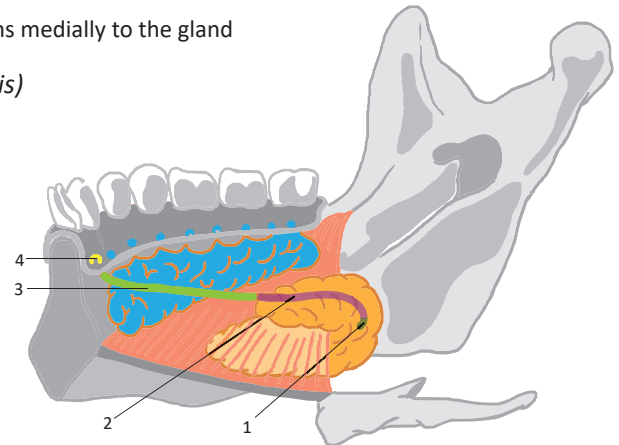
The surface of the gland is covered by the superficial layer of the cervical fascia.

Relationship to the surrounding structures:

- 1 **Superficial cervical ansa** – in contact with the gland through the superficial layer of the cervical fascia
- 2 **Hypoglossal nerve** – in contact with the gland in the inferior pole of the submandibular triangle
- 3 **Facial artery and vein** – cross the gland vertically
– the facial vein is on the outer surface of the gland and the facial artery runs medially to the gland

Course of the submandibular/Wharton's duct (*ductus submandibularis*)

- 1 emerges from the posterior part of the gland
- 2 curves over the posterior margin of the mylohyoid and runs between the mylohyoid and hyoglossus
- 3 continues in the sublingual space on the inner side of the sublingual gland, where it crosses the lingual nerve
- 4 opens on the sublingual caruncle with the major sublingual duct



Sagittal section of the mandible with the sublingual and submandibular gland, viewed from the left

Blood supply and innervation:

Arterial supply: facial artery and lingual artery

Venous drainage: facial vein and lingual vein

Lymphatic drainage: submandibular lymph nodes

Parasympathetic innervation: lingual nerve via the submandibular ganglion (fibres from the facial nerve via the chorda tympani)

Sympathetic innervation: periarterial plexus

Viscerosensory innervation: lingual nerve (from the mandibular nerve)

Parotid gland (*glandula parotidea*)

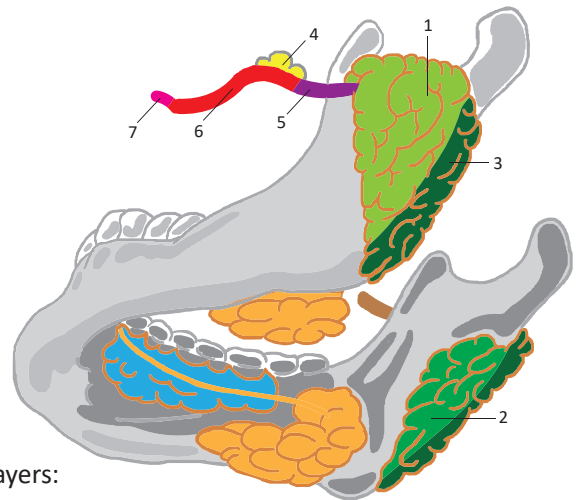
The purely serous parotid gland is the largest salivary gland. It is located on the lateral surface of the head, ventral to the auricle.

Structure and syntopy

- 1 **Superficial part** (*pars superficialis*) – is medially in contact with the masseter
- 2 **Deep part** (*pars profunda*) – is located in the retromandibular fossa
- 3 **Isthmus** – the connection between the superficial and deep part
– is located behind the ramus of the mandible
- 4 **Accessory parotid gland** (*glandula parotidea accessoria*)
– a small accessory gland located above the parotid duct

Parotid duct / Stensen duct (*ductus parotideus*)

- 5 **Masseteric part** (*pars masseterica*) – runs ventrally on the surface of the masseter, about 1cm beneath the zygomatic arch
- 6 **Premasseteric part** (*pars premasseterica*) – courses on the anterior margin of the masseter and enters the buccal fat pad
- 7 **Buccinatoric part** (*pars buccinatoria*) – opens into the oral vestibule on the parotid papilla just above the second upper molar



Mandible and major salivary glands, oblique left posterior view

Parotid fascia (*fascia parotidea*) – covers the gland and consists of three layers:

- 1 **Superficial layer** – part of the parotideomasseteric fascia, covers the masseter
- 2 **Deep layer** – separates the gland from the styloid septum and pterygoids
- 3 **Angular tract** (*tractus angularis*) – an oblique band at the angle of the mandible continues into the superficial layer of cervical fascia
– separates the gland from the submandibular space

Blood supply and innervation:

Arterial supply: superficial temporal artery, maxillary artery, posterior auricular artery, transverse facial artery

Venous drainage: parotid veins (retromandibular vein)

Lymphatic drainage: superficial and deep cervical lymph nodes and parotid lymph nodes

Parasympathetic innervation: auriculotemporal nerve (a branch of the mandibular nerve), carries parasympathetic fibres from the glossopharyngeal nerve

Sympathetic innervation: periarterial plexus

Viscerosensory innervation: auriculotemporal nerve (from the mandibular nerve)

The pharynx is a funnel-shaped muscular organ **located under the base of the skull**. It gets narrow in the craniocaudal direction and is continuous with the oesophagus at the level of vertebra C6. It **communicates ventrally with the nasal, oral and laryngeal cavity**. The oropharynx and laryngopharynx are **part of the digestive and respiratory systems**. The function of the pharynx is provided by two kinds of muscles: **constrictors and levators**. The pharynx plays a role in **swallowing, talking and breathing**.

Parts

- 1 **Nasopharynx** (*pars nasalis pharyngis*) – opens into the nasal cavity through the choanae
- 2 **Oropharynx** (*pars oralis pharyngis*) – opens into the oral cavity through the isthmus of fauces
- 3 **Laryngopharynx** (*pars laryngea pharyngis*) – opens into the larynx through the laryngeal inlet

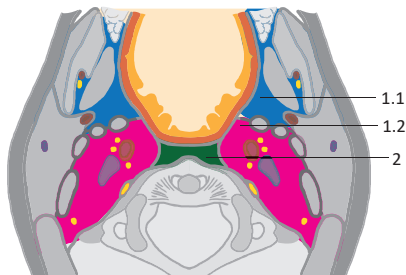
Syntopy

- 4 **Cranially**: cranial base
- 5 **Caudally**: continues as the oesophagus at the level of vertebra C6
- 6 **Ventrally**: opens into the oral, nasal and laryngeal cavities
- 7 **Dorsally**: the cervical part of the vertebral column
- 8 **Laterally**: cervical neurovascular bundle and the lobes of the thyroid gland

Topographical spaces

The space around the pharynx is limited by connective tissue membranes attached to the pharyngeal adventitia.

- 1 **Parapharyngeal space** (*spatium parapharyngeum*)
 - 1.1 **Prestyloid space** (*spatium prestyloideum*)
 - 1.2 **Retrostyloid space** (*spatium retrostyloideum*)
- 2 **Retropharyngeal space** (*spatium retropharyngeum*)



Transverse section of the neck at the level of C1



Sagittal section of the head and neck

Blood supply

Arterial supply: superior thyroid artery, lingual artery and ascending pharyngeal artery (from the external carotid artery), ascending palatine artery (from the facial artery), descending palatine artery and the artery of the pterygoid canal (from maxillary artery)

Venous drainage: pharyngeal venous plexus to the pharyngeal veins (to the internal jugular vein)

Lymphatic drainage: retropharyngeal lymph nodes (to the deep cervical lymph nodes)

Innervation

Pharyngeal plexus – a mixed network of fibres from the glossopharyngeal and vagus nerves and fibres from the superior cervical ganglion of the sympathetic trunk

Visceromotor nerves: vagus nerve

(except the stylopharyngeus, which is innervated by the glossopharyngeal nerve)

Viscerosensory nerves:

Nasopharynx – ventrally: maxillary nerve (from trigeminal nerve), dorsally: glossopharyngeal nerve

Oropharynx – glossopharyngeal nerve and vagus nerve

Laryngopharynx – vagus nerve

Sympathetic system: laryngopharyngeal nerves (from the sympathetic trunk)

Parasympathetic system: facial nerve (nasopharynx)

and vagus nerve (oropharynx and laryngopharynx)

The auditory tube / pharyngotympanic tube / Eustachian tube (*tuba auditiva, tuba pharyngotympanica, salpinx*) is a tube connecting the nasopharynx to the middle ear. It serves to equilibrate air pressure in the middle ear.

The pharyngeal tonsil is also called the adenoid.

The pharyngeal bursa (*bursa pharyngealis*) is an inconstantly present small pit located dorsal to the pharyngeal tonsil. It is a remnant of the embryonic notochord, which becomes enclosed within the pharyngeal wall. This is the first place for lymphatic tissue of the pharynx to appear and can be a place of inflammation.

The pharyngeal hypophysis (*hypophysis pharyngealis*) is a variably present mass of cells located ventral to the pharyngeal bursa. It is a remnant of the embryonic precursor of the adenohypophysis (the pouch of Rathke).

The space of Luschka is a semilunar space around the pharyngobasilar fascia where the pharyngeal wall lacks a muscular layer.

In newborn and infants, the larynx is located more cranially so that the epiglottis overlies the uvula and totally separates the digestive and respiratory systems, enabling concomitant swallowing and breathing.

Lateral bands are regions of lymphoid tissue in the lateral walls of the pharynx.

The pharyngeal recess / fossa of Rosenmüller (*recessus pharyngeus*) is a recess behind the torus tubarius.

Rouviere's lymph node is the most superior retropharyngeal lymph node located below the base of the skull.

Clinical notes

Adenotomy is a surgical removal of the pharyngeal tonsil (adenoid).

A retropharyngeal abscess is a deep circumscribed infection in the neck. This usually results from ascending lymphadenitis after an infection of the upper airways.

The sinus of Morgagni is a weakening in the pharyngeal wall where the auditory tube opens into the pharyngeal cavity. Pathologic processes can spread to the parapharyngeal space and to the base of the skull.

Nasopharynx (*epipharynx, pars nasalis pharyngis*)

Borders

Cranially: cranial base

Caudally: posterior margin of the soft palate

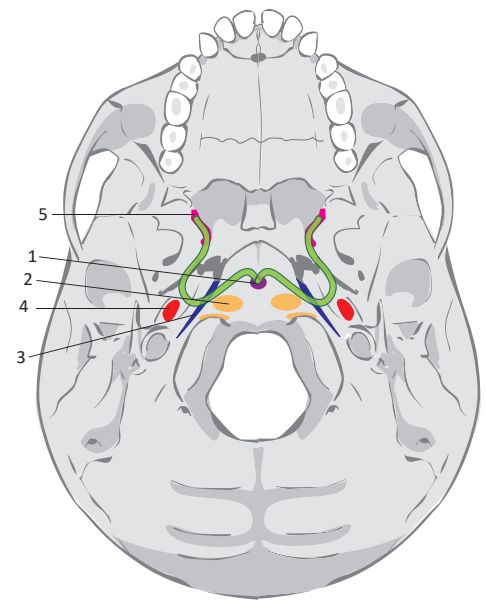
Ventrally: nasal cavity (through choanae)

Insertion of the pharynx on the base of the skull

- the pharynx is inserted on the base of the skull by the pharyngobasilar fascia (*fascia pharyngobasilaris*), a thickening of the submucous connective tissue
- at the level of the pharyngeal insertion on the base of the skull, the pharyngeal wall consists only of connective tissue (pharyngobasilar fascia) without a muscular layer

Detail of the insertion:

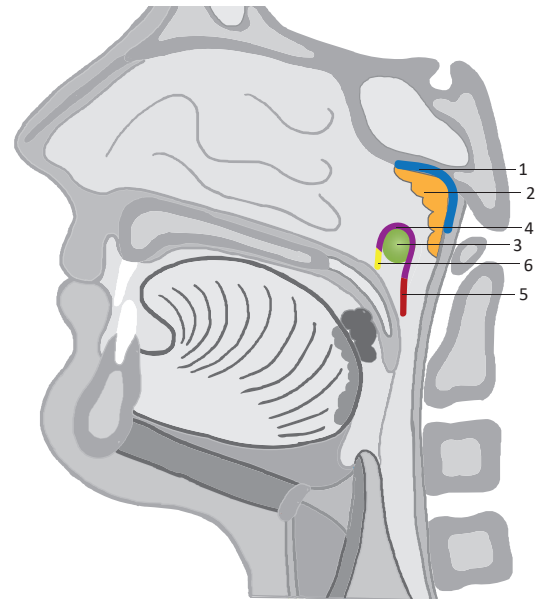
1. from the **pharyngeal tubercle of the occipital bone**
2. further laterally and ventrally to the insertions of the **longus capitis** and **rectus capitis anterior**
3. crosses over the **petro-occipital fissure**
4. ventrally to the **external opening of the carotid canal**
5. directs towards the **medial plate of the pterygoid process of the sphenoidal bone** crossing over on its way the sphenopetrosal fissure



Insertion of the pharynx on the base of the skull

Structures

- 1 **Pharyngeal fornix** (*fornix pharyngis*) – a ventrally opened concave arch that is inserted on the base of the skull
- 2 **Pharyngeal tonsil** / nasopharyngeal tonsil (*tonsilla pharyngealis*) – a mass of lymphoid tissue in the roof of the nasopharynx
- 3 **Pharyngeal orifice of auditory tube** (*ostium pharyngeum tubae auditivae*) – the opening of the auditory tube on the lateral wall of the nasopharynx
- 4 **Torus tubarius** – a mucosal elevation located dorsal to the pharyngeal orifice of the auditory tube, formed by the medial plate of the cartilaginous part of the auditory tube
- 5 **Salpingopharyngeal fold** (*plica salpingopharyngea*) – a mucosal fold descending caudally from the torus tubarius, formed by the salpingopharyngeus
- 6 **Salpingopalatine fold** (*plica salpingopalatina*) – a mucous fold descending ventrocaudally to the soft palate, elevated by homonymous muscle fibres
- 7 **Torus levatorius** – a mucosal bulge between the salpingopharyngeal and salpingopalatine folds under the pharyngeal orifice of the auditory tube, formed by the levator veli palatini
- 8 **Tubal tonsil** (*tonsilla tubaria*) – an accumulation of mucosal lymphoid tissue around the pharyngeal orifice of the auditory tube



Sagittal section of the head and neck

Oropharynx (*pars oralis pharyngis*)

Cranially: the level of the posterior margin of the soft palate (C2 level)

Caudally: the superior margin of the epiglottis (C4 level)

Ventrally: oral cavity (through isthmus of the fauces)

Laryngopharynx (*hypopharynx, pars laryngea pharyngis*)

Cranially: the superior margin of the epiglottis (C4 level)

Caudally: continuous with the oesophagus (C6 level), through the pharyngo-oesophageal constriction

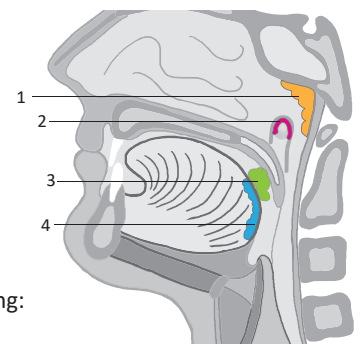
Ventrally: laryngeal cavity (through the laryngeal inlet)

Pharyngeal lymphoid ring (*anulus lymphoideus pharyngis*)

Pharyngeal lymphoid ring / Waldeyer's ring of lymphoid tissue:

The interconnected lymphoid tissue of the pharyngeal mucosa plays an important role in the early recognition of microorganisms entering the upper parts of the digestive and respiratory systems and in the initiation of an immune response. The lymphoid ring is connected caudally to the laryngeal lymphoid tissue. Bands of lymphoid tissue connect neighboring tonsils. Components of the lymphoid ring:

- 1 **Pharyngeal tonsil** – single
- 2 **Tubar tonsil** – paired
- 3 **Palatine tonsil** – paired
- 4 **Lingual tonsil** – single

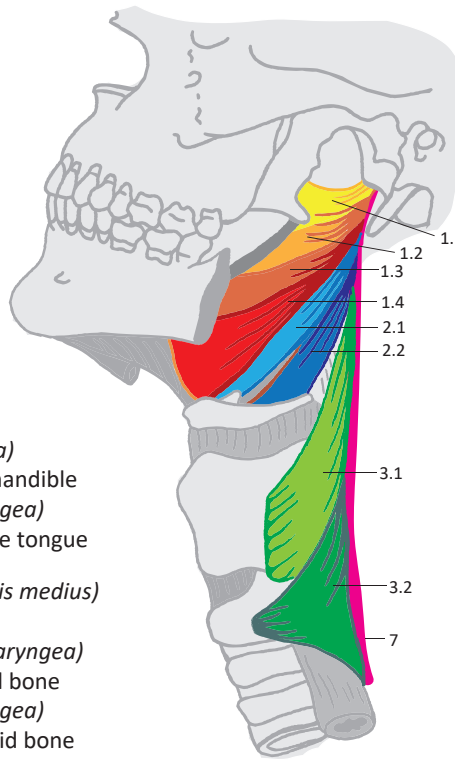


Sagittal section of the head and neck

Pharyngeal constrictors (*musculi constrictores pharyngis*)

The fibres of the pharyngeal constrictors have a **circular course** and **overlap each other in a craniocaudal direction**. They originate on the base of the skull and its neighboring structures and insert into the pharyngeal raphe.

- **7 Pharyngeal raphe** (*raphe pharyngis*) – a band of connective tissue in the posterior wall of the pharynx, where the pharyngeal constrictors are inserted
- **1 Superior constrictor** (*musculus constrictor pharyngis superior*)
 - originates on the base of the skull and its neighboring structures
 - **1.1 Pterygopharyngeal part** (*pars pterygopharyngea*)
 - originates on the medial lamina of the pterygoid process of the sphenoid
 - **1.2 Buccopharyngeal part** (*pars buccopharyngea*)
 - originates on the pterygomandibular raphe
 - **1.3 Mylopharyngeal part** (*pars mylopharyngea*)
 - originates on the mylohyoid line of the mandible
 - **1.4 Glossopharyngeal part** (*pars glossopharyngea*)
 - originates on the transverse muscle of the tongue
- **2 Middle constrictor** (*musculus constrictor pharyngis medius*)
 - originates on the hyoid bone
 - **2.1 Chondropharyngeal part** (*pars chondropharyngea*)
 - originates on the lesser horn of the hyoid bone
 - **2.2 Ceratopharyngeal part** (*pars ceratopharyngea*)
 - originates on the greater horn of the hyoid bone
- **3 Inferior constrictor** (*musculus constrictor pharyngis inferior*)
 - originates on the laryngeal cartilage
 - **3.1 Thyropharyngeal part** (*pars thyropharyngea*)
 - originates on the thyroid cartilage
 - **3.2 Cricopharyngeal part** (*pars cricopharyngea*)
 - originates on the cricoid cartilage

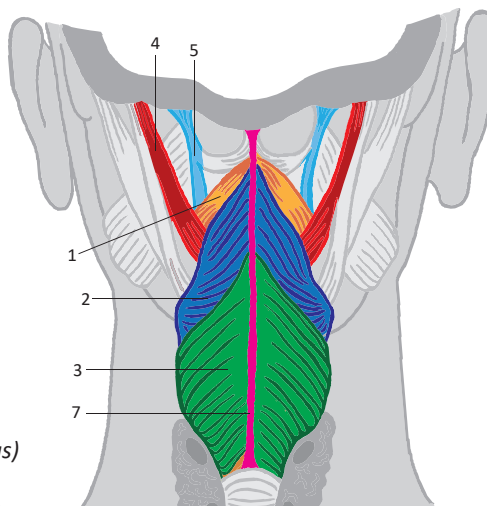


Pharyngeal constrictors, viewed from the left

Pharyngeal levators (*musculi levatores pharyngis*)

The levators have a longitudinal course and descend to the pharyngeal wall from either the outside or the inside of the pharynx. They originate on the base of the skull and its neighboring structures and insert into the pharyngeal muscular wall made by the pharyngeal constrictors.

- **4 Stylopharyngeus** (*musculus stylopharyngeus*)
 - originates on the styloid process and inserts on the outer wall of the pharynx
 - runs through a cleft between the superior and middle constrictors
 - innervation: pharyngeal plexus (n. IX)
- **5 Salpingopharyngeus** (*musculus salpingopharyngeus*)
 - originates on the medial margin of the cartilaginous part of the auditory tube and inserts on the inner wall of the pharynx
 - innervation: pharyngeal plexus
- **6 Palatopharyngeus** (*musculus palatopharyngeus*)
 - originates on the margin of the soft palate, runs in the palatopharyngeal arch and inserts on the inner wall of the pharynx
 - innervation: pharyngeal plexus



Pharyngeal constrictors and levators, posterior view

Oesophagus (or esophagus) is derived from Greek words for carrying (*oesein*) and eating (*phagein*).

The gastro-oesophageal junction / ora serrata / Z-line is the junction between the oesophageal non-keratinised stratified squamous epithelium and the gastric simple columnar epithelium. During endoscopy, this junction is usually well visible.

The interstitial connective tissue between the oesophagus and the deep layer of cervical fascia continues cranially to the retropharyngeal space and caudally to the superior mediastinum. It allows movements of the oesophagus against the vertebral column. However, it also allows pathological processes (mainly infections) to spread towards the base of the skull (in the retropharyngeal space) and to the mediastinum.

Closure of the aboral oesophageal and cardiac orifice relies on the following anatomical structures:

1. **Lower oesophageal sphincter (LES)** – a thickened region of the smooth circular muscle layer between the thoracic and abdominal parts of the oesophagus
2. **Phreno-oesophageal membrane of Leimer**
3. **Crura of diaphragm**
4. **High pressure zone in the muscular layer in the cardia** (approximately 4 mm wide)
5. **Loop of Willis / collar of Helvitius**
6. **Angle of His and valve of Gubaroff**

Clinical notes

Oesophageal varices are a severe consequence of portal hypertension. The elevated portal vein pressure is transmitted to the veins in the submucosa of the cardia and lower part of the oesophagus. Blood stagnates in these veins, resulting in a dilated venous submucosal plexus in cardia and lower third of the oesophagus. There is a risk of the rupture and life threatening bleeding from oesophageal varices.

Metaplasia is a transformation of one type of epithelium to another.

Barrett's oesophagus refers to metaplasia of the normal non-keratinised stratified squamous epithelium lining the lower part of the oesophagus into simple columnar epithelium of the intestinal type. Barrett's oesophagus is caused by gastro-oesophageal reflux and can progress to dysplasia and eventually to oesophageal carcinoma (adenocarcinoma).

The oesophagus is an approximately 25 cm long muscular tube that connects the pharynx to the stomach. Several **curvatures** and **constrictions** are located along **its course through the neck and thorax to the abdominal cavity**. The muscular layer of the upper third of the oesophagus is composed of striated muscle fibres. The middle third is composed of both striated and smooth muscle fibres and the lower third is composed of smooth muscle fibres. Due to the long course of the oesophagus through several topographical regions, the blood supply is provided by multiple sources. The venous system of the oesophagus contains an important **portocaval anastomosis**, which can form varices in portal hypertension.

Parts of the oesophagus and its syntopy

- 1 **Cervical part** (*pars cervicalis*) – runs in the cervical visceral space from C6 to T2
 - ventrally:** trachea
 - dorsally:** the prevertebral layer of the cervical fascia and spinal cord
 - laterally:** cervical neuro-vascular bundle
- 2 **Thoracic part** (*pars thoracica*) – passes through the posterior mediastinum from T2 to T10
 - ventrally:** trachea, main left bronchus, and the heart (left atrium) enclosed in the pericardium
 - laterally:** the vagus nerves run on the sides of the oesophagus
 - running caudally, the right vagus nerve moves to its dorsal surface and the left vagus moves to its anterior surface
 - on the right side:** the azygos vein crosses the right main bronchus and then opens into the superior vena cava
 - on the left side:** the descending part of the aortic arch
 - dorsally:** thoracic aorta and thoracic duct
- 3 **Abdominal part** (*pars abdominalis*) – runs within the peritoneal cavity from T10 to T11
 - a 1–2 cm long portion located in the abdominal cavity after passing through the oesophageal hiatus of the diaphragm
 - ventrally:** the oesophagus is in contact with the liver and creates the oesophageal impression on its dorsal surface
 - dorsally:** diaphragm

Curvatures

In the frontal plane

- the **cervical part** is curved in the median plane
- the **superior thoracic part** is curved to the left
- the **middle thoracic part** is curved to the right
- the **inferior thoracic part** is curved to the left and directs towards the oesophageal hiatus

In the sagittal plane

- the oesophagus is **ventrally concave** in accordance with the kyphosis of the vertebral column

Constrictions

- 4 **Pharyngo-oesophageal constriction of Killian** (*constrictio pharyngoesophagealis*)
 - the border between the pharynx and oesophagus at the level of C6
 - is located between the cricoid cartilage and vertebral column
 - during an endoscopic procedure, this point is about 15 cm from the patient's incisors
- 5 **Thoracic constriction** (*constrictio partis thoracicae, constrictio bronchoaortica*)
 - is in contact with the aortic arch and the main left bronchus at the level of T4–5
 - during an endoscopic procedure, this point is about 25 cm from the patient's incisors
- 6 **Diaphragmatic constriction** (*constrictio phrenica*)
 - is located within the oesophageal hiatus at the level of T10–11
 - during an endoscopic procedure, this point is about 40 cm from the patient's incisors

Blood supply and innervation

Arterial supply is segmental. The individual parts of the oesophagus are supplied from the nearest blood vessels: inferior thyroid artery, oesophageal branches of the aortic arch and thoracic aorta, posterior intercostal arteries, left gastric artery and left inferior phrenic artery.

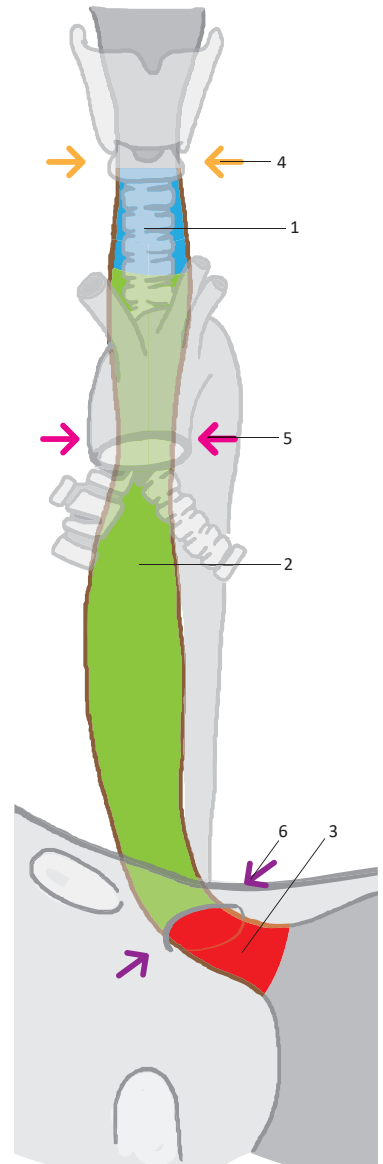
Venous drainage occurs through two venous plexuses located in the wall of the oesophagus: the deep (in the submucosa) and superficial (within the adventitia) plexuses. These plexuses anastomose with each other and form a connection between the riverbeds of the superior vena cava and the portal vein. Thus, they create a portocaval anastomosis. Blood from the oesophagus drains into:

- **superior vena cava:** via the oesophageal veins into the azygos and hemi-azygos veins and also via the inferior thyroid veins
- **portal vein:** the aboral part of the oesophagus via the gastric veins

Lymphatic drainage:

- **cervical part:** deep cervical lymph nodes, **thoracic part:** tracheobronchial lymph nodes and posterior mediastinal lymph nodes, **abdominal part:** left gastric lymph nodes

Mixed innervation: oesophageal plexus (fibres of both vagus nerves and the cervical part of the sympathetic trunk)



Parts and constrictions of the oesophagus

The stomach is the widest part of the digestive tract. It is located in the **supramesocolic part of the peritoneal cavity** under the left vault of the diaphragm. It extends on the right to the epigastric region. The average volume is approximately one litre, but the capacity may be two to three litres. The shape of the stomach varies according to its content and the activity of its muscular wall. The arterial supply is provided by the **coeliac trunk**.

External structure

Parts of the stomach

- **1 Fundus** – the cranial portion under the diaphragm
 - **1.1 Cardial notch** (*incisura cardialis*) – a notch between the fundus and cardia
- **2 Body of stomach** (*corpus gastricum*) – the largest part of the stomach
 - **2.1 Cardia / cardinal part** (*cardia / pars cardica*) – an area around the opening of the oesophagus
 - **2.2 Cardial orifice** (*ostium cardiacum*) – the opening of the oesophagus into the stomach (at the level of T11)
 - **2.3 Gastric canal** (*canalis gastricus*) – a cavity within the body of the stomach
- **3 Pyloric part** (*pars pylorica*) – a horizontal or slightly ascending part of the stomach between the body and duodenum
 - **3.1 Pyloric antrum** (*antrum pyloricum*) – is located between the gastric and pyloric canals at the level of the angular notch
 - **3.2 Pyloric canal** (*canalis pyloricus*) – a 2–3 cm long part heading towards the pylorus
 - **3.3 Pylorus** – the transition between the stomach and duodenum
 - **3.4 Pyloric orifice** (*ostium pyloricum*) – a closable outlet of the stomach into the duodenum

Surfaces

- **4 Anterior wall** (*paries anterior*) – faces the anterior abdominal wall, diaphragm and liver
- **5 Posterior wall** (*paries posterior*) – faces the omental bursa

Curvatures

- **6 Greater curvature** (*curvatura major*) – the left, long and convex curvature of the stomach
 - **6.1 Great notch** (*incisura major*) – a notch within the greater curvature between the fundus and body of the stomach
- **7 Lesser curvature** (*curvatura minor*) – the right, short and concave curvature of the stomach
 - **7.1 Angular notch** (*incisura angularis*) – a small notch within the lesser curvature between the body of the stomach and the pyloric part

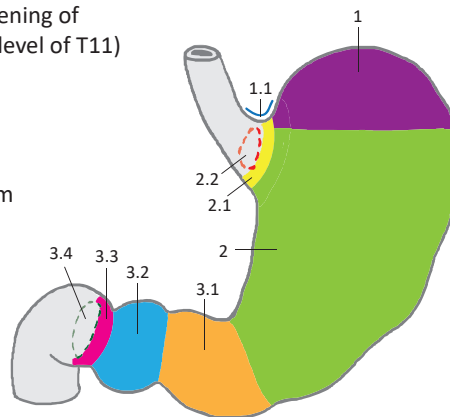
Syntopy

The anterior wall of the stomach is in contact with:

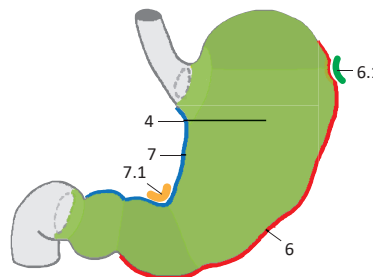
- **1 Diaphragm** – diaphragmatic surface (*facies diaphragmatica*)
- **2 Liver** – hepatic surface (*facies hepatica*)
- **3 Anterior abdominal wall** – free surface (*facies libera*)

Posterior wall of the stomach faces the omental bursa and is in contact with the following structures listed below, through the parietal peritoneum:

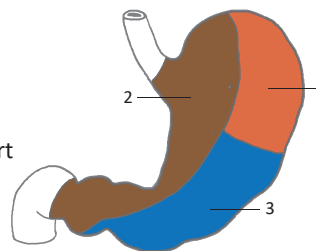
- **1 Diaphragm**
- **2 Left kidney**
- **3 Left suprarenal gland**
- **4 Pancreas**
- **5 Spleen**
- **6 Transverse colon and mesocolon**



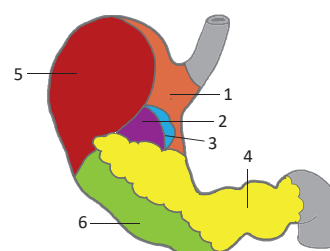
Parts of the stomach



Surfaces and curvatures of the stomach



Syntopy of the anterior wall of the stomach



Syntopy of the posterior wall of the stomach

Stomachus is the Greek term for the stomach. **Ventriculus** is the obsolete Latin term for the stomach.

Functional division of the stomach:

Digestive part (*pars digestoria*) – includes the fundus and body of the stomach.

Evacuating part (*pars egestoria*) – includes the pyloric part of the stomach.

Gastric peristola is a resting phase of the stomach after being filled with food.

Peristaltic waves enable mixing and moving of the gastric content and creation of chyme.

Pyloric pump: peristaltic waves in the pyloric part enable the passage of chyme into the duodenum while the pyloric sphincter is relaxed.

Pepsin, gastrin, intrinsic factor of Castle (necessary for absorption of vitamin B12) and **hydrochloric acid** (HCl) are secreted in the stomach.

The **parasympathetic nervous system** facilitates peristalsis and secretion of HCl.

The **sympathetic nervous system** inhibits peristalsis and secretion of HCl. However, the pyloric sphincter contracts under influence of the sympathetic system.

Mnemonics:

Arteries with a short name (gastric arteries) course along the **lesser curvature**.

Arteries with a long name (gastro-omental arteries) course along the **greater curvature**.

Clinical notes

Pyloric stenosis (*pylorostenosis*) is a congenital hypertrophic stenosis of the pyloric sphincter that may clinically manifest as projectile vomiting.

Achalasia is a pathological motility disorder of the aboral part of the oesophagus. A lack of the myenteric ganglionic cells lead to absence of peristalsis and inability to relax the distal oesophageal sphincter.

The **nodes of Virchow-Troisier** are the left supraclavicular lymph nodes. Enlargement of these nodes can be one of the first symptoms of stomach cancer due to its connection with the thoracic duct.

Gastric ulcers are more likely to occur in areas of the stomach that have poor blood supply, such as where the anastomotic arterial arches are attached to the lesser and greater curvatures of the stomach.

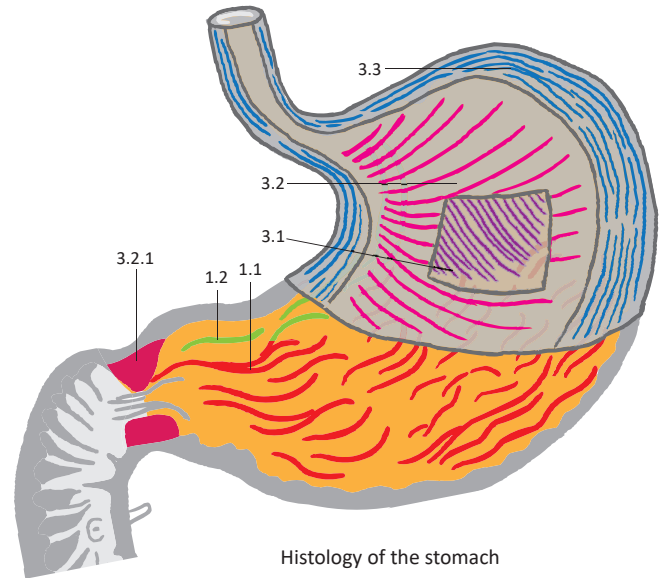
Fixation

Peritoneal duplicatures (ligaments) extend from the serous coat of the stomach and course towards both curvatures. Blood and lymph vessels, nerves and lymph nodes are positioned in the loose fibrous tissue between the sheets of peritoneum.

- 1 **Hepatogastric ligament** (*ligamentum hepatogastricum*) – extends from the lesser curvature to the liver
– part of the lesser omentum
- 2 **Gastrosplenic ligament** (*ligamentum gastrosplenicum*) – extends from the greater curvature to the spleen
- 3 **Gastrophrenic ligament** (*ligamentum gastrophrenicum*) – extends from the greater curvature to the diaphragm
- 4 **Gastrocolic ligament** (*ligamentum gastrocolicum*) – extends from the greater curvature to the transverse colon
– part of the greater omentum

Histology

- 1 **Mucosa** – contains simple columnar epithelium
 - 1.1 **Gastric folds** (*plicae gastricae*) – predominantly longitudinally oriented mucosal folds
– are mainly located along the curvatures
 - 1.2 **Salivary sulcus of Waldeyer** (*sulcus salivarius*)
– longitudinal folds along the lesser curvature
– liquid food may pass through them on their way from the cardia to the pylorus
 - 1.3 **Gastric pits** (*foveolae gastricae*) – gastric glands are located within the lamina propria and open into the gastric pits
 - 1.4 **Gastric areas** (*areae gastricae*)
– areas of mucosa between the gastric pits
- 2 **Submucosa** (*tela submucosa*)
- 3 **Muscular layer** (*tunica muscularis*)
– in addition to the usual circular and longitudinal layers, there is a third innermost obliquely oriented layer
 - 3.1 **Oblique fibres** (*fibrae obliquae*) – the innermost layer
– extend from the circular layer
– course from the cardia to the greater curvature
– their contraction assists in closure of the cardia
 - 3.2 **Circular layer** (*stratum circulare*) – the widest middle circular layer
 - 3.2.1 **Pyloric sphincter** (*musculus sphincter pylori*)
– smooth circular muscle enclosing the pylorus
 - 3.3 **Longitudinal layer** (*stratum longitudinale*) – the outer longitudinal layer
– continuation of closing process of the cardia
– oesophageal longitudinal muscle layer
- 4 **Serosa** (*tunica serosa*) – visceral peritoneum



Histology of the stomach

Blood supply

Arterial supply: coeliac trunk

Anastomosis along the lesser curvature:

- 1 **Left gastric artery** (from the coeliac trunk)
- 2 **Right gastric artery** (usually from the hepatic artery proper)

Anastomosis along the greater curvature:

- 3 **Left gastro-omental (gastro-epiploic) artery** (from the splenic artery)
- 4 **Right gastro-omental (gastro-epiploic) artery** (from the gastroduodenal artery)

Other branches:

- 5 **Short gastric arteries** (from the splenic artery) – to the part of the fundus and body of the stomach
- 6 **Pyloric artery** (from the hepatic artery proper) – to the anterior surface of the pylorus

Venous drainage: portal vein – left and right gastric vein, right and left gastro-omental (gastro-epiploic) vein, short gastric veins
– an important portocaval anastomosis is formed by connections with the oesophageal veins within the area of the cardia
– the ventral surface of the pylorus is drained by the prepyloric vein (either via the right gastric vein or directly to the portal vein)

Lymphatic drainage: coeliac lymph nodes (subsequently to the thoracic duct) and into the left supraclavicular lymph nodes (lymph nodes of Virchow-Troisier)

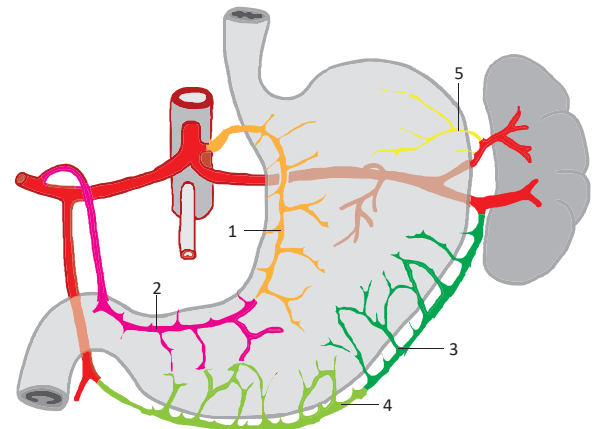
- **body of the stomach:** right and left gastric lymph nodes, right and left gastro-omental lymph nodes
- **fundus:** pancreaticosplenic lymph nodes, **pylorus:** pyloric lymph nodes

Innervation

Parasympathetic system: vagus nerves (anterior and posterior vagal trunk, anterior and posterior gastric branches)

Sympathetic system: splanchnic nerves (fibres of the coeliac plexus and via the periarterial plexuses enter the stomach wall)

Viscerosensory innervation: via the parasympathetic nerves (pressure, cold, heat) and sympathetic nerves (pain)



Blood supply of the stomach

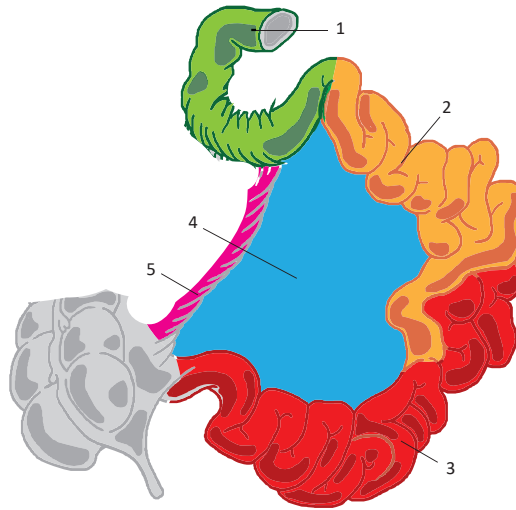
The small intestine is the longest part of the digestive system. It is approximately **3–5 m long** and its width is 2–4 cm. It is composed of **intestinal loops that fill the majority of the abdominal cavity**. The loops face the peritoneum of the anterior abdominal wall and are in contact with parts of the large intestine and other abdominal organs.

Parts

- 1 Duodenum
- 2 Jejunum
- 3 Ileum

Fixation

- 4 Mesentery (*mesenterium*)
 - a peritoneal duplicature suspending the jejunum and ileum
- 5 Root of mesentery (*radix mesenterii*)
 - the attachment of the mesentery to the posterior abdominal wall
 - extends from the duodenojejunal flexure to the right iliac fossa



Histology

Mucosa is covered by simple columnar epithelium and together with the submucosa forms the macroscopically visible transverse circular folds of Kerckring (*plicae circulares*).

The **muscular layer** is composed of a longitudinal and a circular layer. The majority of the intestinal wall is covered by visceral peritoneum, excluding the duodenum, which is mainly retroperitoneal (except its ampulla), with its anterior surface covered by parietal peritoneum.

The term “**duodenum**” refers to its length of 12 inches.

The **jejunum and ileum** are intra-peritoneal organs. They are suspended by the mesentery and, in comparison to the duodenum, which is not suspended by the mesentery, are mobile.

The **muscular layer of the small intestine** performs 3 types of motions. Segmental motions are focal contractions of circular muscle that create segments. Oscillating motions within the segments shorten and lengthen the intestinal loops, mixing the intestinal content. Peristaltic motions move the intestinal content in an aboral direction.

The **intestinal villi** (*villi intestinales*) are prominences of epithelium and lamina propria.

The **intestinal glands / crypts of Lieberkühn** (*glandulae intestinales*) are depressions between bases of the intestinal villi.

Peyer’s patches (*noduli lymphoidei aggregati*) are located in the submucosa of the ileum.

The **intestinal villi of the duodenal wall** are shaped like sheets.

Duodenal glands / Brunner’s glands (*glandulae duodenales*) are submucosal serous tuboalveolar glands that produce and secrete an alkaline liquid, which neutralises the acidic content from the stomach.

The **developmental border zone** between the foregut and midgut is situated just below the major duodenal papilla.

The **suspensory muscle of duodenum / Ligament of Treitz** (*musculus suspensorius duodeni*) is a musculo-ligamentous fascicle extending from the left crus of the diaphragm. It suspends the duodenojejunal flexure.

Clinical notes

Bulbar ulcer is a duodenal ulcer which occurs in the ampulla (duodenal cap). Ulcers are less likely to occur in other parts of the duodenum as chyme gets neutralized by the liquid of the duodenal glands

Oesophagogastroduodenoscopy is an endoscopic examination of the upper parts of the gastrointestinal tract: oesophagus, stomach and duodenum. In the clinical practice, this is often termed gastroscopy.

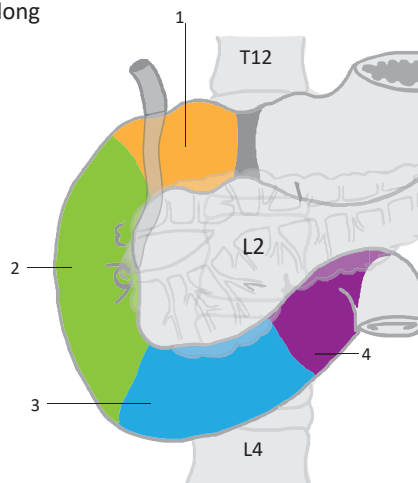
6.1

Duodenum – Duodenum

The duodenum originates at the pyloric orifice of the stomach and continues into the jejunum at the duodenojejunal flexure. It has the shape of a horseshoe. It is located at the level of L1–L3 and its length is about 25–30 cm. The duodenum encircles vertebra L2 from the right side and **forms a space for the head of the pancreas**. The duodenum has four **main parts**, the majority of which are **secondarily retroperitoneal**. The bile duct and both pancreatic ducts open into the duodenum. The duodenal arterial supply is provided by two large trunks from the abdominal aorta: the coeliac trunk and the superior mesenteric artery.

Parts

- 1 **Superior part / D1** (*pars superior*) – is about 4–5 cm long
 - connects to the stomach and is located at the level of vertebra L1
- 2 **Descending part / D2** (*pars descendens*)
 - is about 6–10 cm long
 - is vertically oriented and situated behind the peritoneum between the hilum of the right kidney and the right margin of L2
 - is secondarily retroperitoneal
- 3 **Inferior part / D3** (*pars inferior*)
 - is about 7–8 cm long
 - is horizontally oriented, in front of vertebra L3
 - is secondarily retroperitoneal
- 4 **Ascending part / D4** (*pars ascendens*)
 - is secondarily retroperitoneal
 - runs obliquely up to the left margin of the body of vertebra L2, then continues into the jejunum at the level of the duodenojejunal flexure



Structure

Superior part (*pars superior*)

- 1 **Ampulla / duodenal cap** (*ampulla duodeni*) – an intraperitoneal part connecting the duodenum to the stomach
 - no folds on its mucosa
- 2 **Superior duodenal flexure** (*flexura duodeni superior*) – a flexure between the superior and descending part

Syntopy:

ventrally: gallbladder and the visceral surface of the liver (forms the duodenal impression on the liver)

dorsally: bile duct, portal vein, inferior vena cava, gastroduodenal artery

cranially: hepatoduodenal ligament and its content

caudally: the head of the pancreas

Descending part (*pars descendens*)

- 3 **Longitudinal duodenal fold** (*plica longitudinalis duodeni*)
 - a longitudinal mucosal fold on the medial wall, which is formed by the course of the bile duct
- 4 **Major duodenal papilla of Vater** (*papilla duodeni major*) – an elevated area on the lower end of the longitudinal duodenal fold within the inferior half of the descending part, approximately 10cm from the pylorus
 - the opening of the bile duct and major pancreatic duct
- 5 **Minor duodenal papilla of Santorini** (*papilla duodeni minor*)
 - an elevated spot positioned 2cm above the major duodenal papilla
 - the opening of the accessory pancreatic duct
- 6 **Inferior duodenal flexure** (*flexura duodeni inferior*)
 - a flexure situated between the descending and inferior part

Syntopy:

ventrally: the root of the transverse mesocolon, transverse mesocolon, gallbladder and visceral surface of the liver

dorsally: the bile duct and pancreaticoduodenal arteries in a groove between the duodenum and the head of the pancreas

left: the head of the pancreas and blood vessels (anterior and posterior pancreatic arches)

right: the hilum of the right kidney, right renal pelvis, the initial part of the right ureter and the right renal blood vessels

Inferior (horizontal) part (*pars inferior*)

Syntopy:

ventrally: the root of the mesentery with the superior mesenteric artery and vein (the vein is situated on the right side from the artery), and intestinal loops

dorsally: the right ureter, which crosses the inferior part in its direction from the right to the left, right psoas major, right testicular/ovarian blood vessels, inferior vena cava and abdominal aorta

cranially: the head of the pancreas

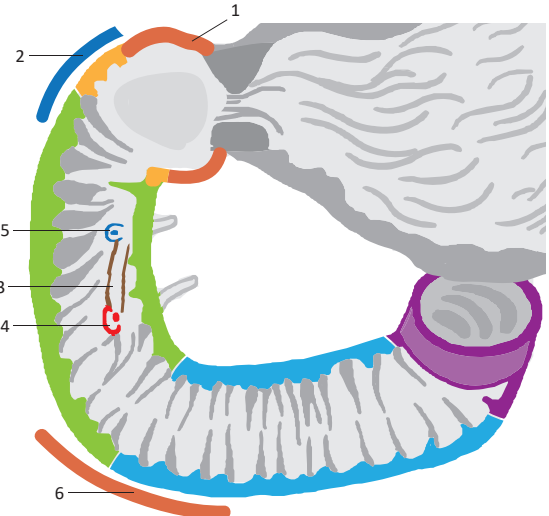
Ascending part (*pars ascendens*)

Syntopy:

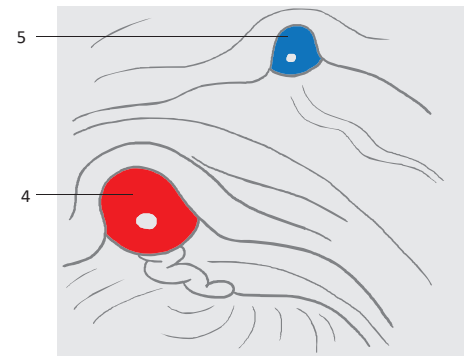
ventrally: intestinal loops, transverse mesocolon, stomach

dorsally: the abdominal aorta, left sympathetic trunk, left psoas major, left testicular/ovarian blood vessels, left renal artery and vein

right: pancreas



Frontal section of the duodenum



Major and minor duodenal papilla, endoscopic inferior view

Blood supply

Arterial blood supply: coeliac trunk and branches form the anterior and posterior pancreatic arches

- 1 **Superior anterior pancreaticoduodenal artery** (from the gastroduodenal artery)
- 2 **Superior posterior pancreaticoduodenal artery** (from the gastroduodenal artery)
- 3 **Inferior pancreaticoduodenal artery** (from the superior mesenteric artery)

Venous drainage: into the hepatic portal vein

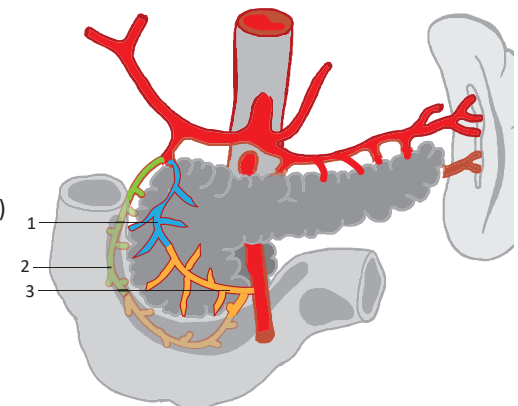
Lymphatic drainage: pyloric, hepatic and superior mesenteric lymph nodes (into the coeliac lymph nodes)

Innervation

Parasympathetic system: vagus nerve (via the coeliac plexus and superior mesenteric periaarterial plexuses)

Sympathetic system: splanchnic nerves (via the coeliac plexus and superior mesenteric periaarterial plexuses)

Viscerosensory innervation: fibres accompanying the vagus and splanchnic nerves



The duodenum becomes the jejunum at the duodenojejunal flexure. However, there is **no macroscopically visible border** between the jejunum and ileum. The jejunum and ileum form loops and together measure about **3 to 5 meters**. Only initial part of the jejunum and the terminal ileum are held firmly in place; the rest of the jejunum and ileum is freely mobile in the abdominal cavity.

Jejunum

- 1 **Diameter:** about 3 cm
- 2 **Length:** two-fifths of the jejuno-ileal length
- 3 **Mucous folds:** tall and numerous
- 4 **Lymphoid nodules:** solitary
- 5 **Arterial supply:** rich (color: rosier in vivo), 1 or 2 series of arcades from the jejunal arteries and long straight arterioles (5 cm)
- 6 **Mesentery:** longer with less fat infiltration
- 7 **Location:** superior and left
- 8 **Autopsy:** no intestinal content

Ileum

- 1 **Diameter:** about 2.5 cm
- 2 **Length:** three-fifths of the total jejuno-ileal length
- 3 **Mucous folds:** short and scarce
- 4 **Lymphoid nodules:** aggregated and solitary
- 5 **Arterial supply:** poor, 2–3 series of arcades from the ileal arteries and short straight arterioles (2 cm)
- 6 **Mesentery:** shorter and with richer fat infiltration
- 7 **Location:** inferior and right
- 8 **Autopsy:** usually presents with some intestinal content

Syntopy

- 1 **Ventrally:** the greater omentum and anterior abdominal wall
- 2 **Dorsally:** the retroperitoneal organs (duodenum, kidneys, ureters, abdominal aorta and inferior vena cava) – the sigmoid colon and rectal ampulla in the pelvic part of the peritoneal cavity,
- 3 **Cranially:** transverse colon
- 4 **Caudally:** organs of the small pelvis
- 5 **Left:** descending and sigmoid colon
- 6 **Right:** caecum and ascending colon

Blood supply

Arterial supply: superior mesenteric artery – jejunal arteries, ileal arteries and ileocolic artery form the mesenteric arterial arches (terminated arcades) – direct branches to the intestinal wall arise from arcades (straight arterioles)

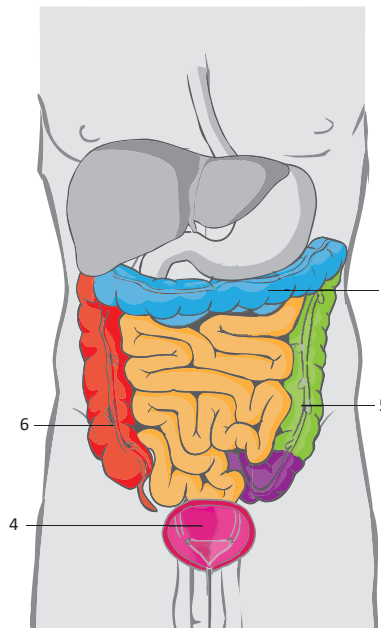
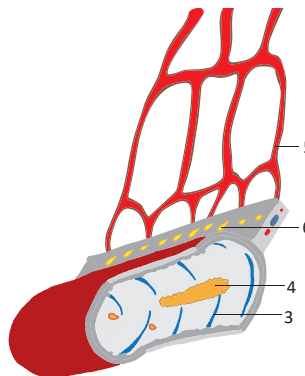
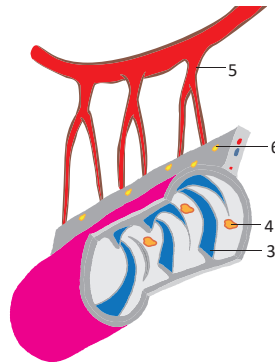
Venous drainage: superior mesenteric vein (into the hepatic portal vein)

Lymphatic drainage: superior mesenteric lymph nodes (into the coeliac and lumbar lymph nodes and subsequently into the thoracic duct)

Innervation

Parasympathetic and sympathetic system: vagus and splanchnic nerves

Viscerosensory innervation: fibers accompanying the vagus and splanchnic nerves



Syntopy of the jejunum and ileum

In the clinical practice the jejunum and ileum are called the jejunoileum or mesenterial intestine.

The terminal ileum is a clinical term for the aboral part of ileum, which continues into the caecum. Inflammatory bowel diseases such as Crohn's disease may affect this part of small intestine.

The enterohepatic circulation refers to the circulation of bile acids that are absorbed in the terminal ileum and then transported by the hepatic portal vein back to the liver to be re-used in bile acid synthesis.

The developmental border zone between the midgut and hindgut is situated between the middle and left thirds of transverse colon (the Cannon-Böhm point). This also represents the border between parasympathetic innervation from the vagus nerve and from the sacral spinal segments.

Gastrocolic ligament (*ligamentum gastrocolicum*) is a part of the greater omentum originating from the greater curvature of the stomach. It is attached to the omental taenia of the transverse colon.

Phrenicocolic ligament (*ligamentum phrenicocolicum*) is a peritoneal fold passing from the left diaphragmatic vault to the left colic flexure.

Clinical notes

Enteroscopy is an endoscopic examination of the small intestine.

The large intestine is **the last part of the digestive tract**. Its main functions are **absorption of water** and creation of faeces. The large intestine is about **1.5 m long** and 6–8 cm wide. Along its course, it is either located secondarily retroperitoneally on the posterior abdominal wall or is suspended in the abdominal cavity by the mesocolon. The large intestine forms a perimeter around the small intestine.

Parts

- 1 Caecum
 - 1.1 Vermiform appendix (*appendix vermiformis*)
- 2 Colon
 - 2.1 Ascending colon (*colon ascendens*)
 - 2.2 Transverse colon (*colon transversum*)
 - 2.3 Descending colon (*colon descendens*)
 - 2.4 Sigmoid colon (*colon sigmoideum*)
- 3 Rectum
 - 3.1 Anus

Histology

Mucosa (and submucosa)

– contains a single layer of columnar epithelium, there are no villi

Muscular layer

– has a circular and a longitudinal layer of smooth muscle
– there are several regions along the course of the large intestine with highly developed circular muscle that form functional sphincters

Superficial layer – is covered by the visceral peritoneum except on the posterior wall of the ascending and descending colon and the aboral part of the rectum, which are connected to the surrounding tissue by adventitia

Appearance

- 1 **Haustra of colon** (*haustra coli*) – saccular pouches of the colon wall are formed by contractions of the layer of circular smooth muscle and bulging of the longitudinal muscles
– notches between the pouches correlate with the internal semilunar folds
- 2 **Taeniae coli** – three longitudinal muscular bands that are visible on the external surface of the colon
– are classified according to the position on the transverse colon:
 - 2.1 **Mesocolic taenia** (*taenia mesocolica*)
– at the attachment of the transverse mesocolon
 - 2.2 **Omental taenia** (*taenia omentalis*) – at the fusion with the greater omentum (gastrocolic ligament)
 - 2.3 **Free taenia** (*taenia libera*)
– faces the anterior abdominal wall
- the taeniae coli form a continuous muscular layer on the **vermiform appendix** and the **rectum** (at the recto-sigmoid transition)
- 3 **Omental appendices / fatty appendices of colon** (*appendices omentales / adiposae coli / appendices epiploicae*)
– small pouches of visceral peritoneum filled with adipose tissue
- 4 **Semilunar folds of colon** (*plicae semilunares coli*)
– transverse semilunar-shaped folds of mucosa

Innervation

Mixed periaortal plexus – extends from the coeliac and superior and inferior mesenteric ganglia

– contains sympathetic, parasympathetic and viscerosensory nerve fibres

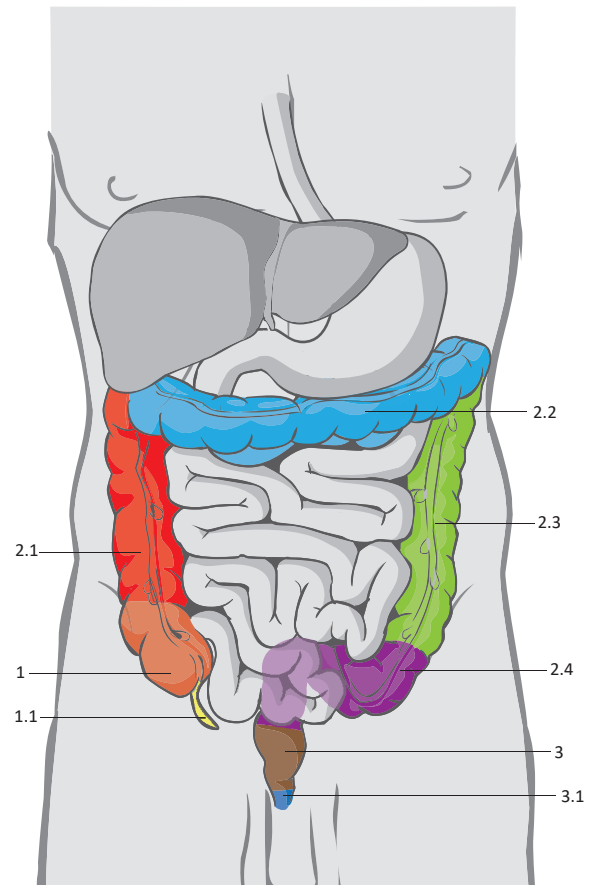
Parasympathetic system: increases peristalsis and contractions of the large intestine and reduces its glandular activity

– **vagus nerve:** innervates the colon until the last third of the transverse colon at the Cannon-Böhm point

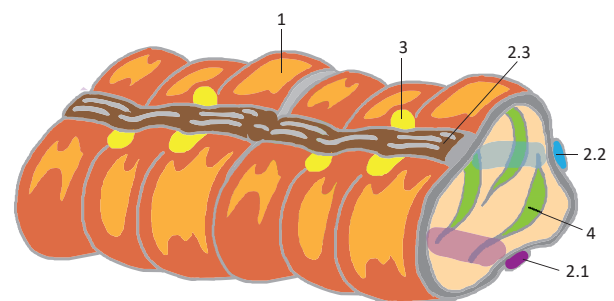
– **sacral parasympathetic fibres:** innervate the colon distal to the Cannon-Böhm point and travel in the sacral splanchnic nerves and the inferior mesenteric plexus

Sympathetic system: reduces peristaltic movements and contractions of the large intestine and increase its glandular activity

Viscerosensory nerves: travel in the vagus nerve and the sacral parasympathetic fibres



Parts of the large intestine



Structure of the wall of the colon

Blood supply

Arterial supply: superior mesenteric artery, inferior mesenteric artery, internal iliac artery

- 1 **Great anastomosis of Haller** – an anastomosis between the left colic and middle colic arteries

Superior mesenteric artery

- 2 **Ileocolic artery** – supplies the terminal ileum and caecum
 - 2.1 **Appendicular artery** – supplies the vermiform appendix
- 3 **Right colic artery** – supplies the ascending colon
- 4 **Middle colic artery** – supplies the right part of the transverse colon

Inferior mesenteric artery

- 5 **Left colic artery** – supplies the left part of the transverse colon and the descending colon
 - 5.1 **Superior branch** – an ascending branch supplying the left part of the transverse colon
 - 5.2 **Inferior branch** – a descending branch that makes an anastomosis with the first sigmoid artery
- 6 **Sigmoid arteries** – 2–4 branches to the sigmoid colon
- 7 **Superior rectal artery** – the terminal branch – supplies the rectal ampulla and anastomoses with the most distal of the sigmoid arteries

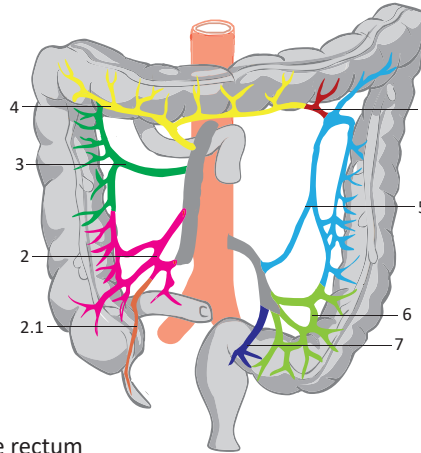
Internal iliac artery

- 8 **Middle rectal artery** – supplies the outer layers of the rectum
- 9 **Inferior rectal artery** – supplies the anal canal and anus – a branch of the internal pudendal artery

Venous drainage: superior rectal vein (to the inferior mesenteric vein and then to the portal vein), middle rectal veins and inferior rectal veins (to the internal iliac vein)

Lymphatic drainage:

- **terminal ileum, caecum, vermiform appendix and beginning of the ascending colon:** ileocolic nodes, precaecal and retrocaecal nodes, appendicular nodes and paracolic nodes
- **ascending colon:** right colic nodes
- **right 2/3 of the transverse colon:** middle colic nodes
- **left 1/3 of the transverse colon, descending colon and sigmoid colon:** left colic nodes
- **rectum:** inferior mesenteric nodes, internal and external iliac nodes, superficial inguinal nodes, pararectal and sacral nodes



Mobile caecum is a term for a caecum, which is fully mobile due to the presence of a fold of peritoneum called the mesocaecum.

Syntopy of the caecum

Ventrally: the greater omentum and ileum

Dorsally: the iliacus, psoas major and lateral cutaneous nerve of the thigh

The appendicular tonsil is a term for the vermiform appendix based on the extensive amount of lymphoid tissue in its wall.

The appendiculo-ovarian ligament of Clado is a variable peritoneal duplicature connecting the vermiform appendix to the right ovary. It contains lymphatic vessels and may allow infection to spread between the vermiform appendix and ovary.

It can be sometimes difficult to determine the position of the vermiform appendix, particularly when it is in atypical positions or during inflammation of the abdominal cavity. In such cases, orientation by the taeniae coli can help as they converge on the vermiform appendix to form a continuous muscular layer.

The right and left paracolic spaces are spaces on the lateral border of the ascending and descending colons. The colon touches the posterior abdominal wall here through the parietal peritoneum.

7.1

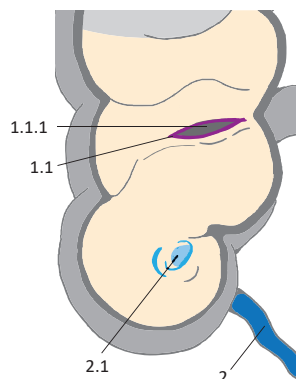
Caecum – Caecum

The caecum is the first part of the large intestine. It is **situated in the right iliac fossa** and it is about 7 cm long. The ileum opens into the left side of the caecum. The caecum is covered by peritoneum, except on the cranial aspect of its dorsal wall. The caudal part of the caecum is free and movable, a peritoneal fold arises here forming the retrocaecal recess. The **vermiform appendix** arises from the caudal apex of the caecum.

External morphology

- 1 **Ileocolic transition** – the junction between the ileum and caecum

- 1.1 **Ileal papilla (papilla ilealis)**
 - 1.1.1 **Ileal orifice** / orifice of the ileal papilla (*ostium ileale caeci*)
 - 1.1.2 **Ileocolic lip** / superior lip (*labrum ileocolicum* / *labium superius*)
 - 1.1.3 **Ileocaecal lip** / Inferior lip (*labrum ileocaecale* / *labium inferius*)
 - 1.1.4 **Frenulum of ileal orifice (frenulum ostii ilealis)** – a small fold of tissue that connects the ileocolic and ileocaecal lips
- 2 **Vermiform appendix (appendix vermiformis)** – a worm-shaped blind-ended tube
 - 2.1 **Orifice of vermiform appendix (ostium appendicis vermiformis)**
 - the opening of the vermiform appendix at the posterior wall of the caecum
 - is located around 3 cm below the ileal orifice

**Clinical notes**

Acute appendicitis is an acute inflammation of the appendix. The classical presentation of appendicitis is pain in the right area under the umbilicus, but it can imitate an inflammation of other organs according to the variability of a position. It is a severe health condition, classified as acute abdomen. Retrocaecal appendices have the worst prognosis.

Atypical positions of the appendix, such as the subhepatic position and cranial displacement during pregnancy, can cause appendicitis to mimic inflammations of other organs, such as the gall bladder and kidneys.

Ulcerative colitis and Crohn's disease belong to a group of diseases called idiopathic inflammatory bowel disease (IBD).

Hirschsprung's disease is a rare congenital disorder. It is characterised by the absence of ganglia in the enteric nervous system in the aboral part of the large intestine.

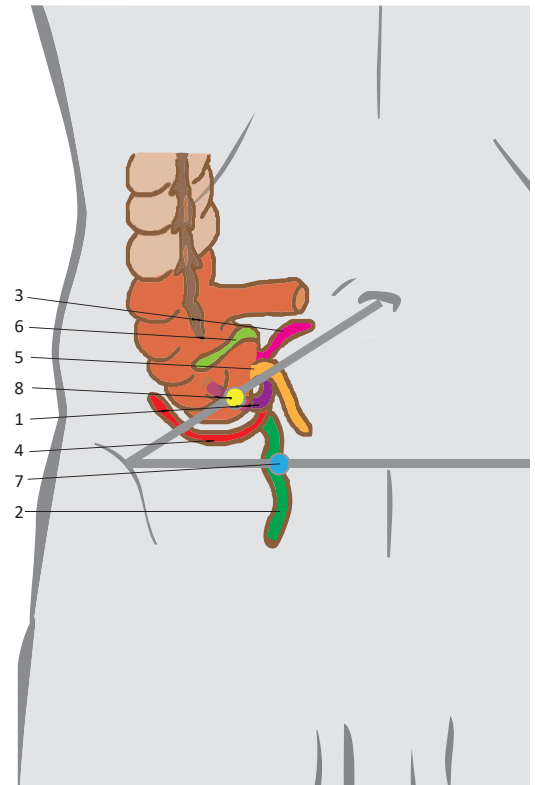
The vermiform appendix is a 5–7 cm long, narrow blind-ended tube. It is held in its location by a **fold of peritoneum called the meso-appendix**. An **abundance of lymphoid tissue** is found throughout its mucosa and submucosa. The taeniae coli converge to form a continuous muscular layer on the vermiform appendix.

Positions of the appendix

- 1 **Retrocaecal position** – the most common position (about 2/3 cases)
 - located behind the caecum, the free part directs cranially
- 2 **Pelvic position** – the second most common position (about 25 % cases)
 - medial to the caecum, crossing the linea terminalis and pointing towards the lesser pelvis
- 3 **Ileocaecal position** – medial to the caecum and parallel to the terminal part of the ileum
- 4 **Laterocaecal position** – laterally to the caecum, between the caecal wall and abdominal wall, the free part points cranially
- 5 **Subcaecal position** – ventrocaudally to the caecum,
 - points towards the inguinal ligament
 - does not cross the linea terminalis
- 6 **Precaecal position** – ventral to the caecum, between the anterior wall of the caecum and abdominal wall

Projections on the anterior abdominal wall

- 7 **Point of Lanz**
 - is located at the border of the right and middle thirds of the interspinous line (a line connecting the superior anterior iliac spine on the right and left sides)
 - corresponds to the pelvic position of the vermiform appendix
- 8 **McBurney's point**
 - about 6 cm from the right superior anterior iliac spine on a line connecting the right superior anterior iliac spine to the umbilicus
 - corresponds to the orifice of the vermiform appendix or to part of its course



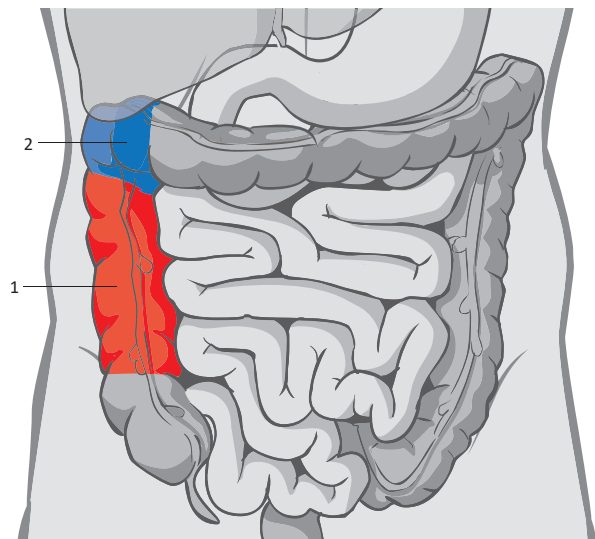
7.2.1

Ascending colon – Colon ascendens

The **ascending colon** is about 15 cm long and is the second part of the large intestine. It is the aboral continuation of the caecum passing on the right side of the abdominal cavity up to the liver. At the liver, it bends to the left, forming the right colic flexure, and continues as the transverse colon. The ascending colon **lacks its peritoneal mesocolon**, because it remains in contact with the posterior abdominal wall and is covered by peritoneum on its ventral aspect. It is a **secondary retroperitoneal organ**.

Syntopy

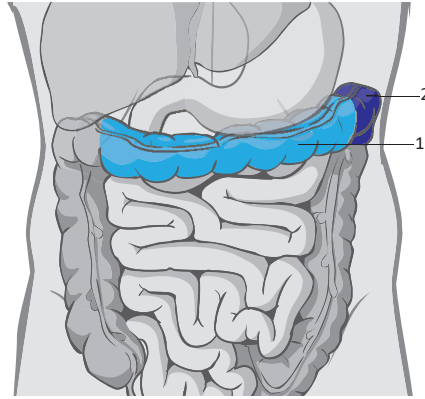
- 1 **Ascending colon** (*colon ascendens*)
 - dorsally:** quadratus lumborum, iliacus, transversus abdominis, inferior pole of the right kidney, iliohypogastric nerve, ilioinguinal nerve, genitofemoral nerve, lateral cutaneous nerve of the thigh
 - medially:** small intestine
 - laterally:** abdominal wall
- 2 **Right colic flexure / hepatic flexure** (*flexura coli dextra/hepatica*)
 - ventrally:** hepatocolic ligament, cystocolic ligament
 - ventromedially:** gallbladder
 - dorsally:** right kidney
 - medially:** descending part of the duodenum
 - cranially:** liver



The **transverse colon** is the continuation of the ascending colon from the **right colic flexure** up to and including the **left colic flexure** at the spleen. At the left colic flexure it turns downwards and continues as the descending colon. The transverse colon is an **intraperitoneal organ**. The **transverse mesocolon** divides the peritoneal cavity into a (upper) **supramesocolic part** and a (lower) **inframesocolic part**.

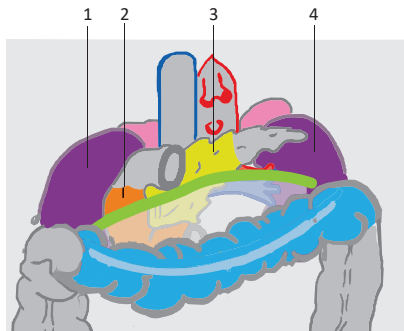
Syntopy

- 1 **Transverse colon** (*colon transversum*)
 - dorsally:** right kidney, duodenum, pancreas, left kidney, spleen
 - ventrally:** greater omentum, anterior abdominal wall
 - cranially:** liver, gallbladder, stomach, spleen
 - caudally:** small intestine
- 2 **Left colic / splenic flexure** (*flexura coli sinistra/splenica*)
 - dorsally:** left kidney
 - ventromedially:** stomach
 - cranially:** spleen



Transverse mesocolon (*mesocolon transversum*)

- **Root of transverse mesocolon** (*radix mesocoli transversi*)
 - the origin of the transverse mesocolon on the parietal peritoneum
 - the root of the transverse mesocolon crosses:
 - 1 **Right kidney**
 - 2 **Descending part of duodenum**
 - 3 **Head and body of pancreas**
 - 4 **Left kidney**



The **root of the mesentery** divides the inframesocolic part of the peritoneal cavity into the left and right paracolic spaces. The superior mesenteric artery supplies the right paracolic space and the inferior mesenteric artery supplies the left paracolic space.

The **haemorrhoidal plexus** (internal rectal venous plexus) is located in the submucosa of the oral part of the anal canal. It elevates the submucosa and contributes to the closure mechanism of the rectum. The superior rectal vein supplies blood to this area and it plays a key role in the pathogenesis of haemorrhoids.

The **connection between the hepatic portal vein and internal iliac vein** is a portocaval anastomosis and can participate in pathogenesis of haemorrhoids.

The **mechanism of defecation** starts with expansion of the rectal ampulla caused by accumulation of stool. This eventually leads to reflex relaxation of the internal anal sphincter. The next step is voluntary relaxation of the external anal sphincter and puborectalis. This triggers contractions of the muscular layer of the rectum that lead to defecation.

The **colorectal zone** reaches aborally down to the supratransitional line.

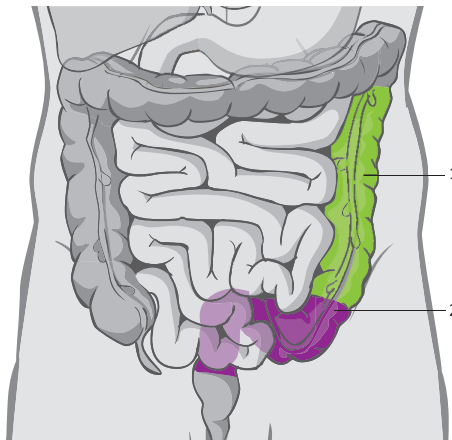
The **surgical anal canal** (described in the text and depicted in the figure) extends from the anorectal junction to the anocutaneous line. The anatomical anal canal extends from the pectinate line to the anocutaneous line.

7.2.3 Descending and sigmoid colon – Colon descendens and sigmoideum

The **descending colon** is 22–30 cm long and passes on the left side of the peritoneal cavity distally from the left colic flexure to the left iliac fossa where it continues as the sigmoid colon. Similarly to the ascending colon, the descending colon **lacks its own peritoneal mesocolon**. It remains in contact with the posterior abdominal wall and is covered by peritoneum only on its ventral aspect. It is a **secondary retroperitoneal organ**. The **sigmoid colon** is 30–40 cm long. It courses from the level of the iliac crest to the level of the vertebral disc S2/S3. It is a primary **intraperitoneal organ** and its **peritoneal duplicature** is called the **sigmoid mesocolon**. It continues as the rectum on the pelvic surface of the sacrum.

Syntopy

- 1 **Descending colon** (*colon descendens*)
 - dorsally:** iliacus, quadratus lumborum, transversus abdominis, left kidney, iliohypogastric nerve, ilioinguinal nerve, genitofemoral nerve, lateral cutaneous nerve of the thigh
 - medially:** small intestine
 - laterally:** abdominal wall
- 2 **Sigmoid colon** (*colon sigmoideum*)
 - in its course it crosses:**
 - left testicular/ovarian vessels
 - left ureter
 - left external iliac vessels



Clinical notes

Structures palpated in the digital rectal examination:

Dorsally: coccyx, pelvic surface of the sacrum

Laterally: ischiadic spines, sometimes also the ovaries in female

Ventrally in a male: prostate (sometimes also the urinary bladder)

Ventrally in a female: the perineal wedge, posterior wall of the vagina and, through the vagina, the cervix of the uterus.

Colonoscopy is an endoscopic examination of the large intestine.

Colorectal carcinoma is one of the most common malignant tumors of the adult population. The common way for metastatic dissemination is hematogenous spread through the hepatic portal vein to the liver and lymphatic spread to the regional lymph nodes.

The rectum is the terminal part of the digestive canal. **It begins at the distal end of the sigmoid colon at the level of the vertebral disc S2/S3 and terminates as the anal canal.** The rectum is about 12–15 cm long, 4 cm wide and is located in the lesser pelvis. In comparison to the other parts of the large intestine, it lacks haustra, taeniae coli and omental appendices. Its shorter oral part has a **peritoneal attachment** called **the mesorectum**. Its longer aboral part is fixed to the surrounding sutures in the lesser pelvis by connective tissue known as **the paraproctium**.

Parts

- 1 **Rectal ampulla** (*ampulla recti*) – the dilated and longer part forming the sacral flexure
 - its mucosa features simple columnar epithelium as so is known as the colorectal zone
 - 1.1 **Anorectal junction** (*junctio anorectalis*) – the transition from the rectal ampulla to the anal canal at the level of pelvic diaphragm
- 2 **Anal canal** (*canalis analis*) – the narrow and shorter part forming the anorectal flexure
 - 2.1 **Supratransitional line** (*linea supratransitionalis*) – a histological border between the colorectal zone and anal transitional zone
 - 2.2 **Anal transition zone ATZ / haemorrhoidal zone** (*zona transitionalis analis*) – the main part of the anal canal
 - the epithelial cells get shorter when going in an aboral direction (from columnar to cuboid and finally squamous)
 - contains the internal rectal venous plexus (haemorrhoidal plexus) in the submucosa
 - 2.2.1 **Anal columns of Morgagni** (*columnae anales*) – 6 to 12 longitudinal folds connected aborally by the anal valves, which form the anal sinuses/crypts, into which the anal glands open
 - 2.3 **Pectinate line / dentate line** (*linea pectinata / linea dentata*) – a border at the level of the anal valves
 - 2.4 **Anal pecten / anoderm** (*zona squamosa analis*) – non-keratinised stratified squamous epithelium without skin adnexa
 - 2.5 **Anocutaneous line / white line of Hilton** (*linea anocutanea*) – the border between the anal pecten and anus
- 3 **Anus** – hole covered by the skin with skin adnexa

Flexures

In the sagittal plane

- 5 **Sacral flexure** (*flexura sacralis*) – the ventrodorsal bend of the rectal ampulla, a dorsal convexity over the sacrum and coccyx
- 6 **Anorectal flexure / perineal flexure** (*flexura anorectalis / flexura perinealis*) – after passing through the pelvic diaphragm, the rectum bends along the coccyx and then courses dorsocaudally

In the frontal plane – flexures correlate with the mucous folds of the rectum – the superior, middle and inferior transverse folds of Houston (*plica transversa recti superior, media et inferior*)

7 **Superodextral lateral flexure** (*flexura superolateralis dextra*) – its convexity directs to the right side

8 **Intermediosinistral lateral flexure / Intermediate lateral flexure** (*flexura intermediolateralis sinistra*) – its convexity directs to the left side

- 8.1 **Middle transverse rectal fold of Kohlrausch** (*plica transversa recti sinistra*) – the biggest fold, about 7 cm from the anal canal

9 **Inferodextral lateral fold / inferior lateral flexure** (*flexura inferolateralis dextra*) – its convexity directs to the right

Muscles

- 10 **Internal anal sphincter** (*musculus sphincter ani internus*) – a smooth muscle located in the oral part of the anal canal
- 11 **External anal sphincter** (*musculus sphincter ani externus*) – a skeletal muscle – attached externally to the internal anal sphincter and from below to the levator ani
- 12 **Puborectalis** (*musculus puborectalis*) – a skeletal muscle, the most medial part of the levator ani – forms a sling around the rectum at the level of the anorectal flexure

Syntopy

Dorsally, in both sexes: sacral vertebrae III–V, coccyx

Laterally, in both sexes: pararectal fossae, levator ani

Ventrally, in male:

- **oral part:** urinary bladder, seminal glands, rectovesical excavation of Proust
- **aboral part:** urinary bladder, seminal glands, ductus deferens, prostate

Ventrally, in female:

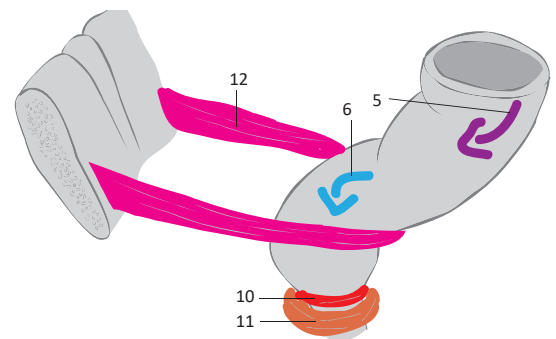
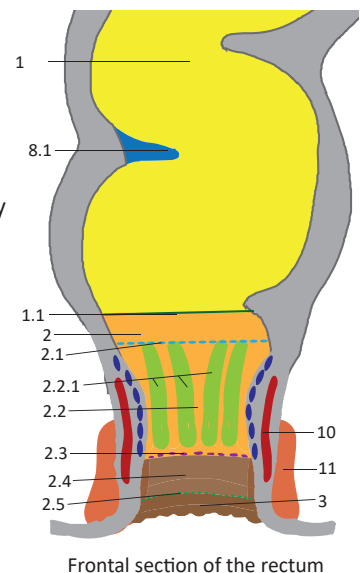
- **oral part:** uterus, vagina, rectouterine excavation of Douglas
- **aboral part:** vagina, perineal muscles

Innervation

Sympathetic: from the lumbar part of the sympathetic trunk via the superior hypogastric plexus (contracts the internal anal sphincter)

Parasympathetic: sacral splanchnic nerves (relaxes the internal anal sphincter and stimulates of the smooth muscles of the rectum)

Somatomotor and somatosensory: pudendal nerves (somatomotor innervation of the external anal sphincter and sensory innervation of the tissue surrounding the anus)



The **liver** is the largest gland in the human body. It has both the exocrine and endocrine functions. The liver **produces bile**, which travels through the biliary tree and is stored in the gallbladder before being excreted by the bile duct into the duodenum. Other important functions of the liver are detoxification and haematopoiesis during foetal development. The liver is **situated in the supramesocolic part of the peritoneal cavity** in the **the right hypochondrium**. A minor part of the liver extends to the left hypochondrium.

Segmentation

Lobes (*lobi*)

– are marked by grooves on the visceral surface

- 1 **Right lobe** (*lobus dexter*) – on the right side of the left sagittal fissure
- 2 **Left lobe** (*lobus sinister*) – on the left side of the left sagittal fissure
- 3 **Quadrate lobe** (*lobus quadratus*) – below the porta hepatis
- 4 **Caudate lobe** (*lobus caudatus*) – above the porta hepatis

Segments (*segmenta*)

– are divided by the branching pattern of the blood vessels and biliary tree
– the inner segmentation does not correlate with the pattern of lobes as seen from the surface

- I **Posterior segment** – caudate lobe (*segmentum posterius*)
- II **Left posterior lateral segment** (*segmentum posterius laterale sinistrum*)
- III **Left anterior lateral segment** (*segmentum anterius laterale sinistrum*)
- IV **Left medial segment** (*segmentum mediale sinistrum*)
- V **Right anterior medial segment** (*segmentum anterius mediale dextrum*)
- VI **Right anterior lateral segment** (*segmentum anterius laterale dextrum*)
- VII **Right posterior lateral segment** (*segmentum posterius laterale dextrum*)
- VIII **Right posterior medial segment** (*segmentum posterius mediale dextrum*)

Histology

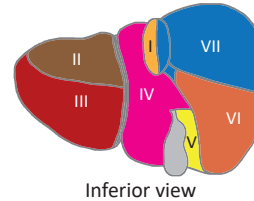
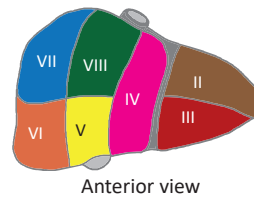
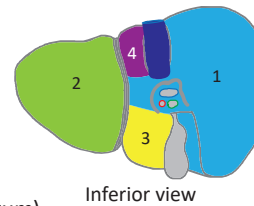
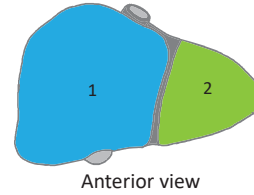
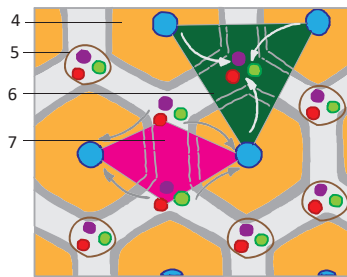
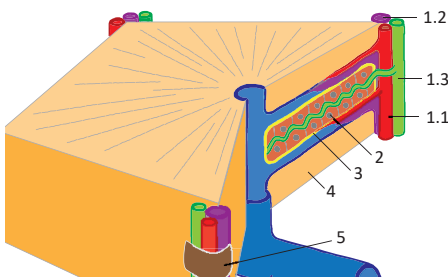
1 Portal triad of Glisson (*trias hepatica*)

– consists of the following structures, running between the lobules:

- 1.1 **Interlobular artery** (*arteria interlobularis*) – a branch of the hepatic artery proper
- 1.2 **Interlobular vein** (*vena interlobularis*) – a branch of the hepatic portal vein
- 1.3 **Interlobular bile duct** (*ductus bilifer interlobularis*) – a branch of the biliary tree

Hepatic parenchyma / hepatic muralium (*muralium hepaticum*) – tissue made of hepatocytes

- 2 **Hepatocyte** – a hepatic cell, the main cell of the liver parenchyma
- 3 **Hepatocyte lamina/plate/trabecula** – a radiating column made of hepatocytes
- 4 **Classical/polygonal hepatic lobule** – the basic anatomical unit
– a hexagonal shape, the hepatocyte laminae radiating outward from the central vein
- 5 **Portobiliary space** – the point of the contact of three hepatic lobules, contains the portal triad
- 6 **Portal lobule of Mall** – the principal unit of circulation, a triangular shape, vertices are represented by three central veins, the portobiliary space is located in the centre
– a part of the hepatic parenchyma supplied by the interlobular vessels located in its centre
- 7 **Hepatic/liver acinus** – the primary functional unit
– made of hepatic laminae supplied by one particular circumlobular vessels (artery and vein)
– quadrilateral (two triangles), the circumlobular artery and vein are situated in its center, vertices are represented by two central veins and two portobiliary spaces



The basic structural unit of the liver: hepatocytes

The basic morphologic unit: the classical hepatic lobule

The basic circulation unit: the portal lobule

The basic functional unit: the hepatic acinus

The fibrous capsule of the liver (capsule of Glisson) is a fibrous cover of the liver. It projects septa into the liver parenchyma. These septa delimit individual hepatic lobules and house the blood vessels and bile ducts (perivascular fibrous capsule).

The sinusoid is a type of a blood capillary which does not have a continuous basal lamina. Sinusoids have irregularly shaped lumens and walls, which contain fenestrations and pores. Sinusoids are located between the anastomosing laminae of the hepatic parenchyma. Their structure facilitates exchange of substances between blood and hepatocytes. The portal blood and arterial blood are mixed in the sinusoids (and also in the circumlobular veins and precapillary veins).

The round ligament of the liver is a ligamentous remnant of the umbilical vein.

The ligamentum venosum is a ligamentous remnant of the venous duct.

The hepatorenal ligament is a transition between the dorsal part of the right coronary ligament and the parietal peritoneum covering the right kidney.

The fibrous appendix of the liver (*appendix fibrosa hepatis*) is the end of the left coronary ligament attached to the diaphragm.

Portal vein page 298.

Clinical notes

Liver cirrhosis is conversion of the hepatic parenchyma into nodules. Portal venous blood pressure may rise due to cirrhosis. This is called portal hypertension and can lead to an expansion of the portocaval anastomosis between the stomach and oesophagus, which results in oesophageal varices. Bleeding of ruptured varices is a life-threatening condition.

Division of the liver parenchyma into segments allows surgical removal of particular segments according to their blood supply.

External structure

Surfaces

- **1 Diaphragmatic surface** (*facies diaphragmatica*) – faces the diaphragm
 - **1.1 Bare area** (*area nuda*) – a triangular area devoid of peritoneum
 - adheres to the diaphragm
- **2 Visceral surface** (*facies visceralis*) – in contact with abdominal organs
- **3 Inferior border** (*margo inferior*) – the border of the diaphragmatic and visceral surfaces

Grooves – divide the visceral surface into lobes

- **4 Left sagittal fissure** (*fissura sagittalis sinistra*) – is divided into a ventral and dorsal part
 - **4.1 Fissure for round ligament** (*fissura ligamenti teretis*) – the ventral part of the left sagittal fissure, contains the round ligament of the liver
 - **4.2 Fissure for ligamentum venosum** (*fissura ligamenti venosi*)
 - the dorsal part of left sagittal fissure, contains the ligamentum venosum
- **5 Right sagittal fissure** (*fissura sagittalis dextra*)
 - **5.1 Fossa for gallbladder** (*fossa vesicae biliaris/felleae*) – a groove for the gallbladder
 - **5.1.1 Gallbladder** (*vesica biliaris/fellea*)
 - **5.2 Groove for vena cava** (*sulcus venae cavae*) – the dorsal part of the right sagittal fissure
 - contains the inferior vena cava, at the entrance of the hepatic veins
 - **5.2.1 Ligament of vena cava** (*ligamentum venae cavae*)
 - a ligament holding the inferior vena cava to the liver
- **6 Porta hepatis** (*porta hepatis*) – a transverse groove, where the hepatic ducts emerge from the liver and vessels and nerves enter the liver
 - **6.1 Right and left branch of the hepatic artery proper**
 - **6.2 Right and left branch of the hepatic portal vein**
 - **6.3 Right and left hepatic duct**
 - **6.4 Lymphatic vessels**

Fixation

The liver is held to the surrounding structures by peritoneal duplicatures and, in some places, by single sheets of peritoneum.

- **7 Falciform ligament** (*ligamentum falciforme hepatis*) – a peritoneal duplicature
 - is located in front of the liver between the right and left lobes
 - attaches the liver to the anterior abdominal wall
 - splits into two separate sheets on the bare area, the anterior sheet continues as the coronary ligament on both sides
- **8 Round ligament of liver** (*ligamentum teres hepatis*)
 - a remnant of the fetal umbilical vein in the free edge of the falciform ligament extending from the umbilicus to the liver
- **9 Coronary ligament** (*ligamentum coronarium hepatis*) – a single sheet of peritoneum
 - borders the superior and posterior parts of the diaphragmatic surface
 - **9.1 Right coronary ligament** (*ligamentum coronarium dextrum*)
 - borders the bare area on the right side (both ventrally and dorsally)
 - **9.1.1 Right triangular ligament** (*ligamentum triangulare dextrum*)
 - a continuation to the right side as a peritoneal duplicature
 - **9.2 Left coronary ligament** (*ligamentum coronarium sinistrum*)
 - borders the bare area on the left side (both ventrally and dorsally)
 - **9.2.1 Left triangular ligament** (*ligamentum triangulare sinistrum*)
 - a continuation to the left side as a peritoneal duplicature
 - **10 Lesser omentum** (*omentum minus*) – a peritoneal duplicature between the oesophagus, stomach, duodenum and liver
 - **11 Hepato-oesophageal, hepatogastric and hepatoduodenal ligament**

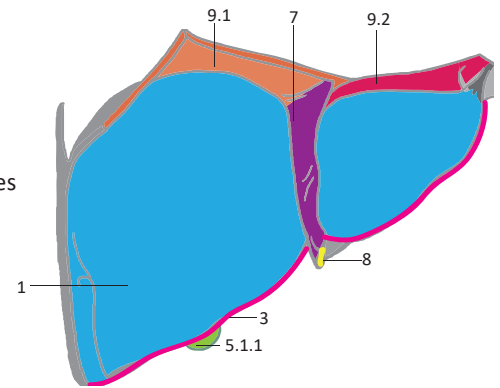
Syntopy

Impressions on the visceral surface

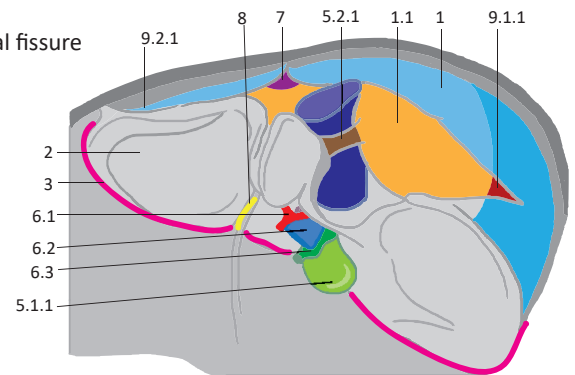
- **1 Gastric impression** (*impressio gastrica*) – an impression of the anterior wall of the stomach
- **2 Renal and suprarenal impressions** (*impressio renalis et suprarenalis*) – impressions of the right kidney and right suprarenal gland
- **3 Duodenal impression** (*impressio duodenalis*) – an impression of the ampulla of the duodenum
- **4 Colic impression** (*impressio colica*) – an impression of the right colic flexure
- **5 Oesophageal impression** (*impressio oesophagealis*) – an impression of the abdominal part of the oesophagus

Impressions on the diaphragmatic surface

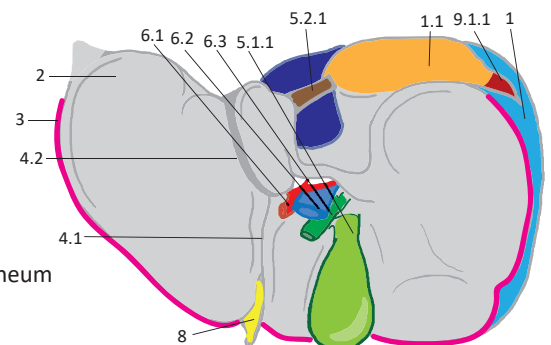
- **6 Cardiac impression** (*impressio cardiaca*) – an impression of the heart in the middle of the diaphragmatic surface
 - the heart exerts pressure on left hepatic lobe, between the heart and left hepatic lobe is the pericardium and diaphragm



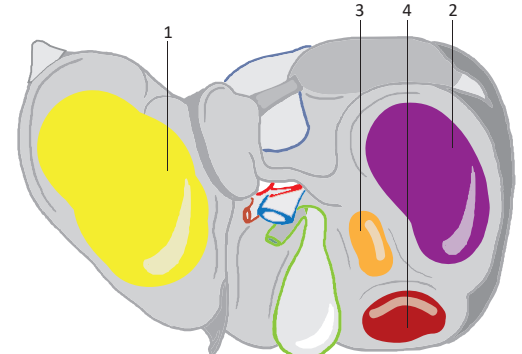
Anterior view of the diaphragmatic surface



Posterior view of the visceral surface



Inferior view of the visceral surface



Inferior view of the visceral surface

Intrahepatic bile ducts

Intrahepatic bile ducts – the ducts collecting bile inside the liver

1. **Bile canaliculi / bile capillaries** (*canaliculi biliferi*) – hepatocytes produce bile and form the walls of the bile canaliculi, which flow into intralobular bile ducts
2. **Intralobular bile ducts** (*canales biliferi*) – at the periphery of the hepatic lobule the bile canaliculi transform into the short cholangiocyte-lined intralobular bile ducts
3. **Interlobular bile ducts** (*ductus biliferi interlobulares*) – are located in the portobiliary space, empty into the segmental ducts, which empty into the lobar ducts
4. **Right and left hepatic ducts** (*ductus hepaticus dexter et sinister*) – emerge from the liver at the porta hepatis

Blood supply

Functional circulation (*vasa publica*)

1. blood from the abdominal organs enters the liver via the hepatic portal vein
2. to the interlobular veins, which, at right angles, give off the circumlobular veins (*venae circumlobulares*) that surround the hepatic lobule and empty into the hepatic sinusoids via the precapillary veins
3. blood in the sinusoids supplies hepatocytes with oxygen and nutrients
4. and flows to the central vein, sublobular veins (*venae sublobulares*), hepatic veins and finally to the inferior vena cava

Nutritive circulation (*vasa privata*)

1. blood flows from the hepatic artery proper
2. to the interlobular arteries, which give off the circumlobular arteries at right angles that flows via arterioles to the terminal parts of the hepatic portal system
3. the nutritive system oxygenates the wall of the veins and interstitial connective tissue and partially oxygenate hepatocytes (hepatocytes are mainly supplied by the portal venous blood)

Arterial supply: from the coeliac trunk

– proper hepatic artery (branch from the common hepatic artery)

Portal venous supply: from the hepatic portal vein

– the hepatic portal vein is formed by the superior mesenteric vein and splenic vein

Venous drainage: to the inferior vena cava

- right, left and intermediate hepatic veins
- usually two venous roots from the right side and one from the left side

Lymphatic drainage:

- to the hepatic and coeliac nodes
- also through the diaphragm to:
 - anterior and posterior mediastinal nodes,
 - anterior and posterior phrenic nodes, parasternal nodes

Innervation

Viscerosensory: right phrenic nerve

Parasympathetic: vagus nerve

– decreases glycogen and protein catabolism and increases production of bile

Sympathetic: thoracic splanchnic nerves

– decreases production of bile and increases glycogen and protein catabolism

Autonomic nerve fibers: supply the liver and gallbladder in a mixed hepatic plexus, passing via the coeliac ganglia and winding around the proper hepatic artery

Functions

1. **Bile production**
2. **Metabolism** of saccharides, lipids and amino acids
3. **Detoxification** of ammonia, alcohol and drugs
4. **Degradation** of hormones, heme and cholesterol
5. **Storage** of vitamins and glycogen
6. **Haematopoiesis** during fetal development

Cholékystis is the Greek term for gallbladder.

The mucosa of the gallbladder has columnar epithelium and displays macroscopic folding.

The spiral fold is a mucosal fold in the cystic duct, which allows bile to flow in both directions.

Hartmann's pouch is an out-pocketing of the gallbladder wall situated at the border between the neck of the gallbladder and first part of the cystic duct.

The sphincter of the ampulla is closed during the resting phase of digestion. This causes bile to accumulate in the bile duct and, after it reaches the level of the cystic duct, passively flow into the gallbladder.

The cystohepatic triangle of Calot is a topographical space bordered by the liver, common hepatic and cystic duct. It is a site for ligation of the cystic artery during cholecystectomy.

The liver is held in its position by the inferior vena cava, its adhesion to the diaphragm (bare area), support of the abdominal organs, the round ligament of the liver and other attachments.

The gallbladder projects on the anterior abdominal wall at Murphy's point, which is located at the intersection of the midclavicular line and costal margin.

The major duodenal papilla projects on the anterior abdominal wall at Desjardin's pancreatic point, which is located 6 cm from the umbilicus on a line connecting the umbilicus to the right axilla.

Clinical notes

Murphy's point may be sensitive on palpation (Murphy's sign). This indicates pathology of the gallbladder.

Cholelithiasis is the presence of bile stones in the gallbladder (cholecystolithiasis) or in the bile ducts (choledocholithiasis). This can cause inflammation of the gallbladder (cholecystitis), bile ducts (choledocholithitis) or pain, caused by the stone moving through the bile ducts (biliary colic).

Endoscopic retrograde cholangiopancreatography (ERCP) is an endoscopic technique that allows visualisation of the biliary tree and pancreas. It allows contrast material can be injected into the bile duct and pancreatic duct.

The **gallbladder** is an about 10 cm long blind sac with a volume of about 15 ml. It is located **in the fossa for the gallbladder on the visceral (inferior) surface of the liver**. **Bile is collected and concentrated** in the gallbladder. Almost the whole gallbladder is covered by the visceral peritoneum except part of its anterior wall, which is attached to the liver by connective tissue (there can be sometimes a short peritoneal duplication). The bile duct leads bile to the duodenum and to the gallbladder. Two types of the bile ducts exist: intrahepatic (inside the liver) and extrahepatic (outside the liver) bile ducts.

Structure

- 1 **Fundus** – located beyond the level of the inferior margin of the liver
- 2 **Body (corpus)** – located in the fossa for the gallbladder
- 3 **Infundibulum** – the part where the body continues to the neck
- 4 **Neck (collum)** – continues to the cystic duct

Syntopy

ventrally: the visceral surface of the liver

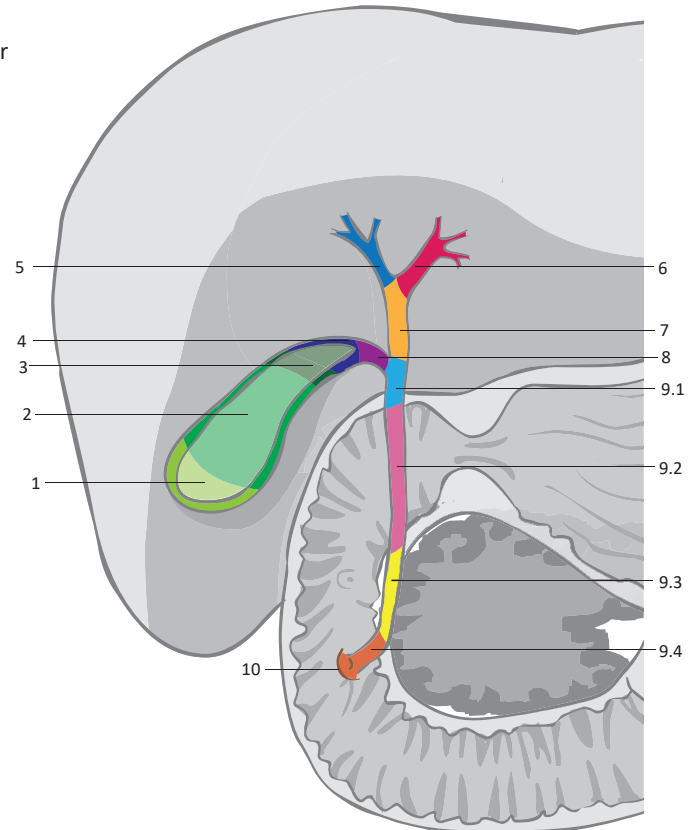
– the fundus is in contact with the anterior abdominal wall

dorsally: right colic flexure, transverse colon

Extrahepatic bile ducts

– course from the porta hepatis to the duodenum within the hepatoduodenal ligament

- 5 **Right hepatic duct** (*ductus hepaticus dexter*)
 - located on the right side of the porta hepatis
- 6 **Left hepatic duct** (*ductus hepaticus sinister*)
 - located on the left side of the porta hepatis
- 7 **Common hepatic duct** (*ductus hepaticus communis*)
 - formed by communion of the right and left hepatic ducts
- 8 **Cystic duct** (*ductus cysticus*) – connects the gallbladder to the common hepatic duct on the right side at a distance of about 4 cm from the porta hepatis
- 9 **Bile duct** (*ductus choledochus*) – formed by the common hepatic duct and cystic duct
 - is about 6–8 cm long and is divided into 4 parts:
 - 9.1 **Supraduodenal part** (*pars supraduodenalis*)
 - above the superior part of the duodenum
 - 9.2 **Retroduodenal part** (*pars retroduodenalis*)
 - behind the superior part of the duodenum
 - 9.3 **Pancreatic part** (*pars pancreaticae*)
 - located in a groove between the descending part of the duodenum and head of the pancreas
 - elevates the longitudinal fold of the duodenum (*plica longitudinalis duodeni*)
 - 9.4 **Intramural part** (*pars intramuralis*)
 - passes inside the medial wall of the descending part of the duodenum



Gallbladder and bile ducts

Bile ducts termination

- 10 **Major duodenal papilla of Vater** (*papilla duodeni major*) – a mucous elevation located at the inferior border of the longitudinal fold of the duodenum in the lower half of the descending part of the duodenum, about 10 cm aboral to the pylorus
 - drains the bile duct and pancreatic duct
- 11 **Hepatopancreatic ampulla** (*ampulla hepatopancreatica*) – an enlargement at the union of the bile duct and pancreatic duct
- 12 **Sphincter of ampulla** (*musculus sphincter ampullae hepatopancreaticae*) – a sphincter surrounding the hepatopancreatic ampulla
- 13 **Sphincter of the bile duct / Sphincter of Oddi** (*musculus sphincter ductus choledochi*) – a sphincter at the distal end of the bile duct

Blood supply and innervation

Arterial supply: cystic artery (a branch of the right branch of the hepatic artery proper)

Venous drainage: cystic vein (to the portal vein)

Parasympathetic: vagus nerve (stimulates contractions of the gallbladder and relaxes the sphincter of the bile duct)

Sympathetic: thoracic splanchnic nerves (decrease contractions of the gallbladder and stimulate contraction of the sphincter of bile duct)

Viscerosensory: right phrenic nerve

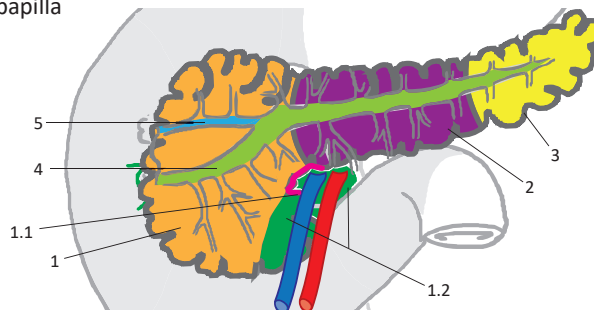
The **pancreas** is a **secondary retroperitoneal** glandular organ located behind the stomach and in front of the inferior vena cava and abdominal aorta. It has both exocrine and endocrine functions. The **exocrine part** produces pancreatic juice that contains a variety of enzymes. Pancreatic juice flows into the duodenum. The **endocrine part** releases hormones into the blood. One of the important hormones it releases is **insulin**.

Structures

- **1 Head of pancreas** (*caput pancreatis*) – the large and rounded part of the pancreas
 - is located in front of the body of vertebra L2 within the concavity of the duodenum
 - **1.1 Pancreatic notch** (*incisura pancreatis*)
 - an notch on the inferior aspect of the pancreas between its head and body
 - **1.2 Uncinate process** (*processus uncinatus*) – a extension of the head of the pancreas
 - passes behind the superior mesenteric vessels
- **2 Body of pancreas** (*corpus pancreatis*) – the narrower and longer part of the pancreas
 - courses horizontally to the left side over the large vessels
 - 2.1 **Omental eminence** (*tuber omentale*)
 - a ventral prominence elevated by the abdominal aorta
- **3 Tail of pancreas** (*cauda pancreatis*) – points towards the spleen, located intraperitoneally

Ducts

- **4 Pancreatic duct / duct of Wirsung** (*ductus pancreaticus*)
 - about 10–12 cm long, runs along the whole length of the pancreas
 - runs dorsocaudally inside the head of the pancreas and emerges usually together with the bile duct (the hepatopancreatic ampulla) into the descending part of the duodenum on the major duodenal papilla
 - 4.1 **Sphincter of pancreatic duct** (*musculus sphincter ductus pancreatici*)
 - a sphincter at the distal end of the pancreatic duct
- **5 Accessory pancreatic duct / duct of Santorini** (*ductus pancreaticus accessorius*)
 - is usually connected to the pancreatic duct but drains independently into the minor duodenal papilla



Pancreas located in the duodenal concavity

Syntopy

Head of pancreas

dorsally: bile duct, hepatic portal vein, inferior vena cava

ventrally: the root of the transverse mesocolon (via the omental bursa)

Body of pancreas

dorsally: abdominal aorta, superior mesenteric vessels, left renal vessels, left suprarenal gland

ventrally: stomach (via omental bursa)

cranially: on the upper border courses tortuous splenic artery (splenic vein runs behind the pancreas)

caudally: transverse colon, duodenojejunal flexure, jejunum

Blood supply and innervation

Arterial supply: coeliac trunk and superior mesenteric artery

- branches form anastomoses with each other
- the anterior and posterior superior pancreaticoduodenal artery (branches of the gastroduodenal artery) and the inferior pancreaticoduodenal artery (a branch of the superior mesenteric artery) supply the head of the pancreas
- pancreatic branches of the splenic artery (dorsal pancreatic artery, greater pancreatic artery, caudal pancreatic artery, inferior pancreatic artery) supply the body and tail of the pancreas

Venous drainage: to the splenic vein, superior mesenteric vein and then to the portal vein

Lymphatic drainage: hepatic, coeliac and pancreaticosplenic nodes

Sympathetic, parasympathetic and viscerosensory: pancreatic plexus (from the coeliac plexus)

The **exocrine part of the pancreas** is composed of the pancreatic acini which are made up by the serous cells. It is the major part of the pancreas.

The **endocrine part of the pancreas** is composed of about one million pancreatic islets of Langerhans. It represents only 1.5 % of the total pancreatic mass.

The **duodenal window** is a radiological term for the position of the head of the pancreas in the concavity of the duodenum at the level of L2.

The **pancreas originates** as two buds which merge during prenatal development. The ventral bud forms the inferior part of the head of the pancreas and its uncinate process and the dorsal bud form the superior part of the head, the body and the tail of the pancreas.

Divided pancreas (*pancreas divisum*) is a developmental defect presenting with a missing connection between the pancreatic duct and the ducts draining the body and tail. In this condition, most of the pancreatic juice is drained via the accessory pancreatic duct.

Anular pancreas (*pancreas anulare*) is a developmental defect characterised by a pancreas circumferentially surrounding the descending part of the duodenum. It can lead to constriction of the duodenal lumen.

Peritoneum consists the mesothelium supported by subserous connective tissue, which attaches the peritoneum to the surrounding structures. It contains blood and lymphatic vessels.

Mesothelium is a layer of simple squamous epithelium that produces peritoneal fluid. Mesothelial cells have the ability to absorb fluid.

The **total area of the peritoneum** is approximately 2 m².

Clinical notes

Tumors of the head of the pancreas can compress the bile duct leading to icterus (jaundice), compress the portal vein leading to portal hypertension and compress the pylorus or duodenum leading to obstruction of the passage of chyme.

Peritoneum is a **serous membrane** lining the abdominal cavity and part of the pelvic cavity. The **peritoneal cavity** is a space between the parietal and visceral peritoneum. The **parietal peritoneum** covers the abdominal walls and the **visceral peritoneum** covers organs. **Peritoneal duplicatures** are folds of peritoneum connecting the visceral to the parietal peritoneum. They **hold organs in their correct positions** and allow **blood vessels and nerves to pass to the organs**.

Principal parts

- 1 **Parietal peritoneum** (*peritoneum parietale*)
- 2 **Visceral peritoneum** (*peritoneum viscerale*)
- 3 **Peritoneal cavity** (*cavitas peritonealis*)
- 4 **Mesentery** (*mesenterium*) – a peritoneal duplicature
- 5 **Root of mesentery** (*radix mesenterii*)
– the attachment of the duplicature to the posterior abdominal wall

Intraperitoneal organs and their duplicatures

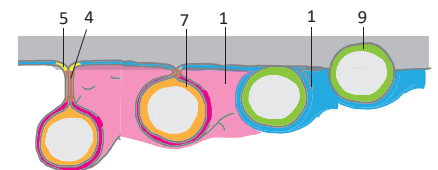
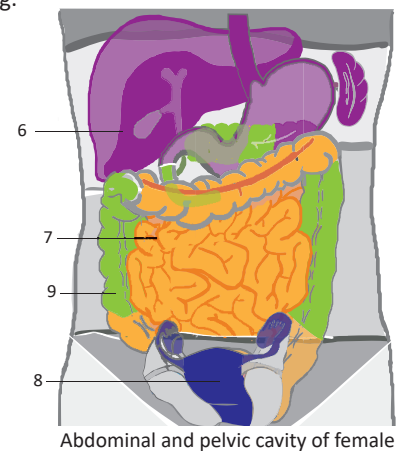
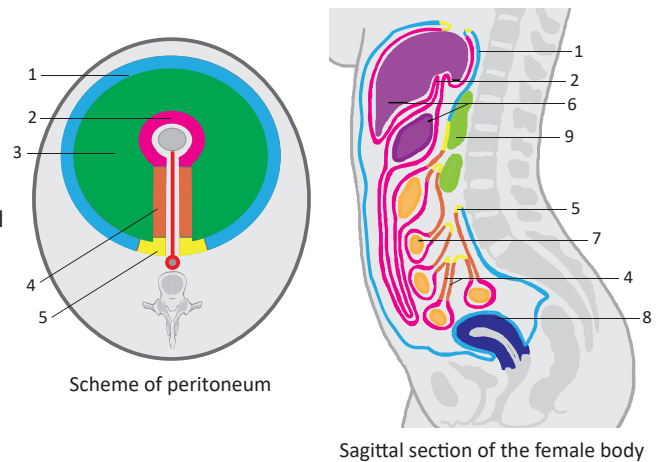
- 6 **Supramesocolic part of peritoneal cavity** (*pars supramesocolica cavitatis peritonealis*)
 - 6.1 **Stomach** – gastrophrenic ligament, gastrosplenic ligament, gastrocolic ligament, hepatogastric ligament
 - 6.2 **Oesophagus** – hepato-oesophageal ligament
 - 6.3 **Spleen** – phrenicosplenic ligament
 - 6.4 **Liver and gallbladder** – falciform ligament, coronary and triangular ligament, hepatorenal lig.
 - 6.5 **Pancreas** (tail only)
 - 6.6 **Duodenum** (ampulla only) – hepatoduodenal ligament, duodenorenal ligament
- 7 **Inframesocolic part of peritoneal cavity** (*pars inframesocolica cavitatis peritonealis*)
 - 7.1 **Jejunum and ileum** – mesentery (*mesenterium*)
 - 7.2 **Caecum and vermiform appendix** – mesocaecum and mesoappendix
 - 7.3 **Transverse colon** – transverse mesocolon
 - 7.4 **Sigmoid colon** – sigmoid mesocolon
 - 7.5 **Rectum** (oral part only) – mesorectum
- 8 **Pelvic peritoneal cavity** (*cavitas peritonealis pelvis*)
 - 8.1 **Uterus and uterine tube** – mesometrium and mesosalpinx
 - 8.2 **Ovary** – mesovarium

Secondary retroperitoneal organs and their origin

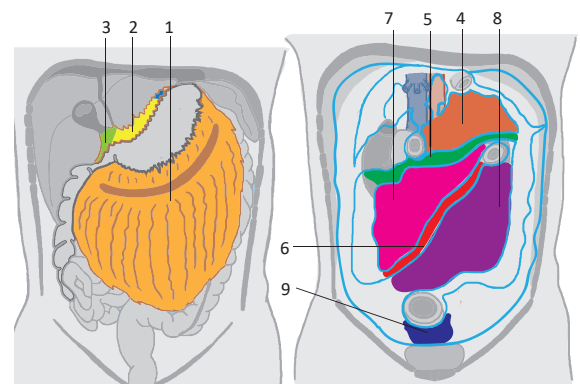
- 9 **Secondary retroperitoneal organs** – organs primary formed intraperitoneally but lose their peritoneal duplicature during development as they move to the retroperitoneal space
 - 9.1 **Ascending and descending colon** – only their anterior wall is covered by the visceral peritoneum
 - 9.2 **Pancreas** – except the tail
 - 9.3 **Duodenum** – except the ampulla

Principal structures and parts of the peritoneal cavity

- 1 **Greater omentum** (*omentum majus*) – a peritoneal duplicature extended from the greater curvature of the stomach to the transverse colon, it hangs in front of the small intestine
- 2 **Lesser omentum** (*omentum minus*) – a peritoneal duplicature extended between the liver and the oesophagus, stomach (lesser curvature) and duodenum
- 3 **Hepatoduodenal ligament** (*ligamentum hepatoduodenale*) – the right margin of the lesser omentum containing the following structures: portal vein, proper hepatic artery, bile duct
- 4 **Omental bursa / lesser sac** (*bursa omentalis*) – the space behind the stomach in the supramesocolic part of the peritoneal cavity
- 5 **Root of transverse mesocolon** (*radix mesocoli transversi*)
– an attachment of the transverse mesocolon which divides the peritoneal cavity into two parts – supramesocolic and inframesocolic part
- 6 **Root of mesentery** (*radix mesenterii*) – attachment of the peritoneal duplicature for the jejunum and ileum (mesentery)
- 7 **Right paracolic space** – the right part of the inframesocolic part
- 8 **Left paracolic space** – the left part of the inframesocolic part, communicates with the pelvic peritoneal cavity
- 9 **Rectouterine pouch of Douglas** (*excavatio rectouterina*) / **Rectovesical pouch of Proust** (*excavatio rectovesicalis*)
– the lowest part of the peritoneal cavity in female/male



Origin of the secondary retroperitoneal organs



Greater omentum, anterior view

Parietal peritoneum on the posterior abdominal wall, anterior view

I. General overview of the digestive system

1. State the four principal parts of the digestive tract. (p. 171)

II. Mouth

2. State and explain the deciduous and permanent dental formulae. (p. 175)
3. Explain the innervation of the tongue. (p. 177)
4. State the topographic structures and spaces located lateral to the palatine tonsil. (p. 178)
5. List the 5 muscles of the soft palate and fauces and give their innervation. (p. 179)
6. List the three major salivary glands and describe the topography of their ducts. (p. 180–181)

III. Pharynx

7. Describe the insertion of the pharynx on the skull. (p. 183)
8. List the three sinuses which communicate with the lumen of the pharynx. (p. 182)
9. List the constrictors and levators of the pharynx and their innervation. (p. 184)

IV. Oesophagus

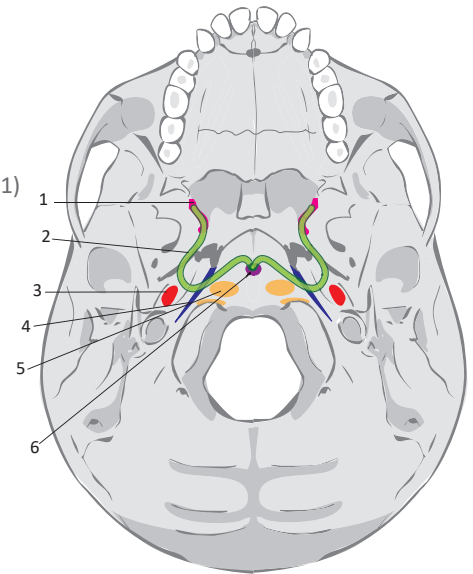
10. Explain the three main narrowings of the oesophagus. (p. 185)
11. Explain the term “oesophageal varices”. (p. 184)
12. Describe the relationship between the oesophagus and trachea, and the oesophagus and pericardium. (p. 185)
13. State the venous plexuses located the oesophagus. (p. 185)

V. Stomach

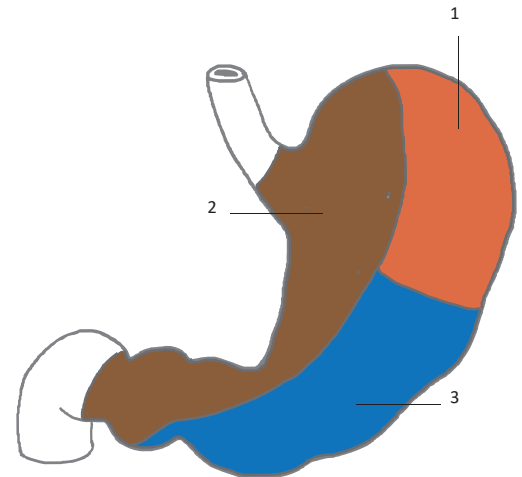
14. State the organs, which are in contact with the posterior wall of the stomach. (p. 186)
15. List the 4 arteries, which form anastomotic arches on the curvatures of the stomach. (p. 187)
16. Explain the anatomical principle of the closure of the sphincters between the stomach and oesophagus and between the stomach and duodenum. (p. 184, p. 186)
17. Describe the arterial supply of the stomach. (p. 187)

VI. Small intestine

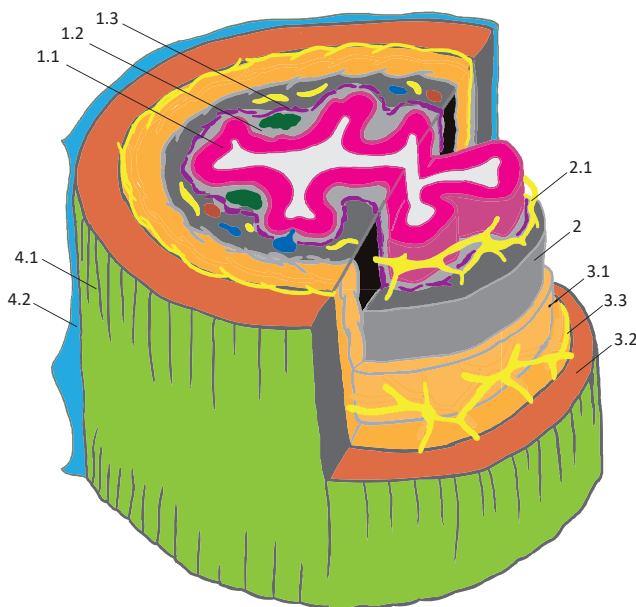
18. List 4 structures located dorsal to the superior part of the duodenum. (p. 189)
19. Describe the topography and content of the root of the mesentery. (p. 188)
20. Define the principal morphological differences between the jejunum and ileum. (p. 190)
21. State the principal arteries that supply the duodenum. (p. 189)



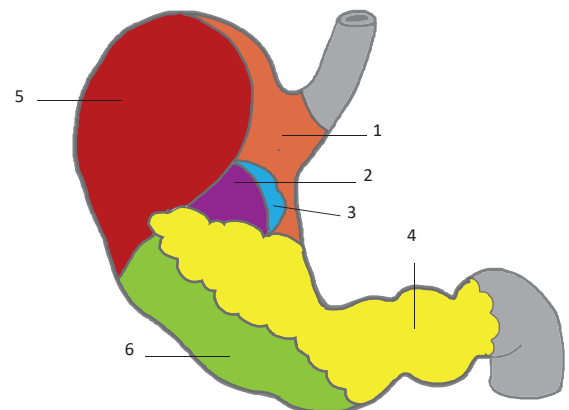
Describe the insertion of the pharynx on the cranial base



Describe the syntopy of the anterior wall of the stomach



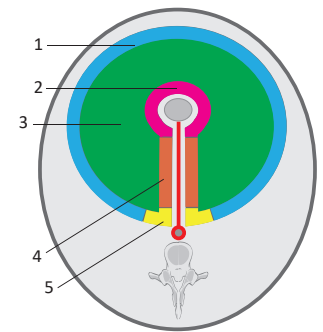
Describe the histology of the digestive tract



Describe the syntopy of the posterior wall of the stomach

VII. Large intestine

22. Describe the course of the root of the transverse mesocolon. (p. 194)
23. State 2 structures, which are located dorsal to the rectum. (p. 195)
24. List the structures and spaces that are located ventral to the rectum in the male. (p. 195)
25. Name the structures and spaces which are located ventral to the rectum in the female. (p. 195)
26. Define the most common positions of the vermiform appendix. (p. 193)
27. Describe the points of projection of the vermiform appendix on the anterior abdominal wall. (p. 193)
28. State the branches of the superior mesenteric artery supplying the large intestine. (p. 192)
29. State the branches of the inferior mesenteric artery supplying the large intestine. (p. 192)
30. State the principal arterial branches supplying the rectum. (p. 192)



Describe the scheme of the peritoneum

VIII. Liver

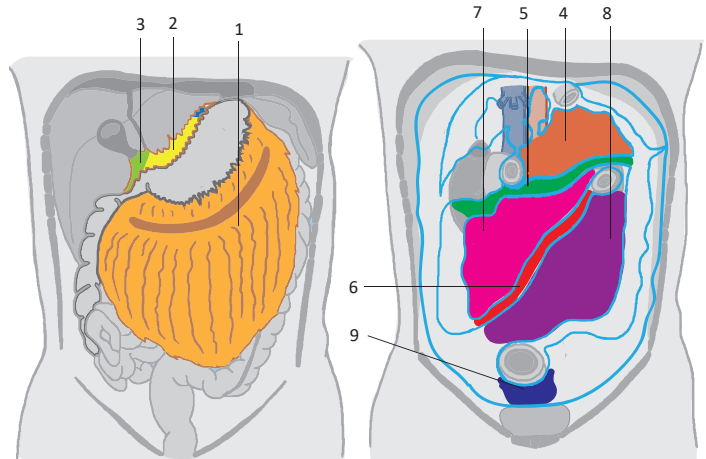
31. State 5 impressions of organs on the visceral surface of the liver. (p. 197)
32. Explain the term "portal triad". (p. 196)
33. Explain the difference between the coronary and triangular ligaments. (p. 197)
34. Describe the two types of blood circulation in the liver. (p. 198)

IX. Gallbladder

35. List the 5 parts of the extrahepatic bile ducts. (p. 199)
36. List the 4 parts of the bile duct. (p. 199)
37. Explain the abbreviation „ERCP“. (p. 198)

X. Pancreas

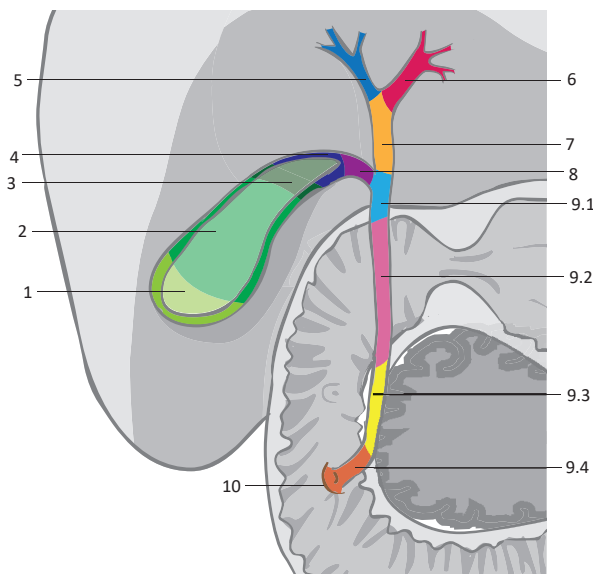
38. Explain the topographic relationship between the pancreas and duodenum. (p. 188, 200)
39. Name the ducts of the pancreas and describe how and where they open. (p. 200)
40. List the branches of the coeliac trunk that supply the pancreas. (p. 200)
41. List the branches of the superior mesenteric artery that supply the pancreas. (p. 200)
42. List the branches of the splenic artery that supply the pancreas. (p. 200)



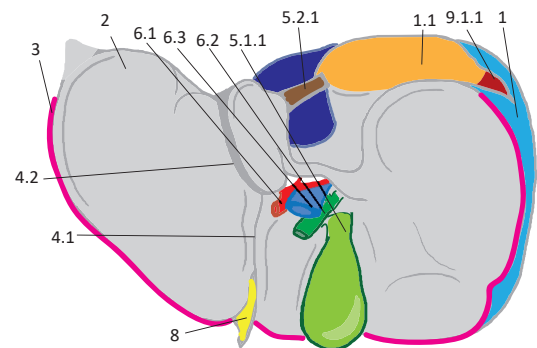
Describe the principal structures and parts of the peritoneal cavity

XI. Peritoneum

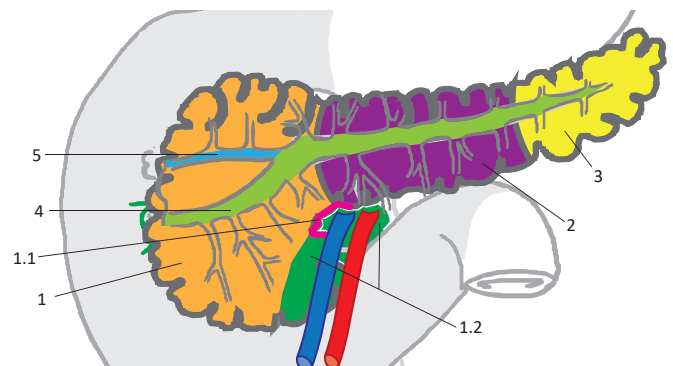
43. Explain the terms "intraperitoneal", "primary retroperitoneal" and "secondary retroperitoneal" (p. 201)
44. State 4 secondary retroperitoneal organs. (p. 201)
45. Describe the histology of the peritoneum. (p. 200)



Describe the parts of the gallbladder and the extrahepatic bile ducts



Describe the visceral surface of the liver



Describe the principal parts of the pancreas and its relationship to the duodenum

We would like to **thank the following anatomists, physicians and medical students** for their invaluable help, devotion and feedback in the preparation of this chapter.

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